A First Course in PROBABILITY

EIGHTH EDITION



SHELDON ROSS

CHAPTER 4

Random Variables

4.1 RANDOM VARIABLES
4.2 DISCRETE RANDOM VARIABLES
4.3 EXPECTED VALUE
4.4 EXPECTATION OF A FUNCTION OF A RANDOM VARIABLE
4.5 VARIANCE
4.6 THE BERNOULLI AND BINOMIAL RANDOM VARIABLES
4.7 THE POISSON RANDOM VARIABLE
4.8 OTHER DISCRETE PROBABILITY DISTRIBUTIONS
4.9 EXPECTED VALUE OF SUMS OF RANDOM VARIABLES
4.10 PROPERTIES OF THE CUMULATIVE DISTRIBUTION FUNCTION

4.1 RANDOM VARIABLES

Frequently, when an experiment is performed, we are interested mainly in some function of the outcome as opposed to the actual outcome itself. For instance, in tossing dice, we are often interested in the sum of the two dice and are not really concerned about the separate values of each die. That is, we may be interested in knowing that the sum is 7 and may not be concerned over whether the actual outcome was (1, 6), (2, 5), (3, 4), (4, 3), (5, 2), or (6, 1). Also, in flipping a coin, we may be interested in the total number of heads that occur and not care at all about the actual head–tail sequence that results. These quantities of interest, or, more formally, these real-valued functions defined on the sample space, are known as *random variables*.

Because the value of a random variable is determined by the outcome of the experiment, we may assign probabilities to the possible values of the random variable.

EXAMPLE 1a

Suppose that our experiment consists of tossing 3 fair coins. If we let Y denote the number of heads that appear, then Y is a random variable taking on one of the values 0, 1, 2, and 3 with respective probabilities

$$P\{Y = 0\} = P\{(T, T, T)\} = \frac{1}{8}$$

$$P\{Y = 1\} = P\{(T, T, H), (T, H, T), (H, T, T)\} = \frac{3}{8}$$

$$P\{Y = 2\} = P\{(T, H, H), (H, T, H), (H, H, T)\} = \frac{3}{8}$$

$$P\{Y = 3\} = P\{(H, H, H)\} = \frac{1}{8}$$

Since Y must take on one of the values 0 through 3, we must have

$$1 = P\left(\bigcup_{i=0}^{3}\{Y = i\}\right) = \sum_{i=0}^{3} P\{Y = i\}$$

which, of course, is in accord with the preceding probabilities. ■

EXAMPLE 1b

Three balls are to be randomly selected without replacement from an urn containing 20 balls numbered 1 through 20. If we bet that at least one of the balls that are drawn has a number as large as or larger than 17, what is the probability that we win the bet?

Solution. Let X denote the largest number selected. Then X is a random variable taking on one of the values $3, 4, \ldots, 20$. Furthermore, if we suppose that each of the $\binom{20}{3}$ possible selections are equally likely to occur, then

$$P\{X = i\} = \frac{\binom{i-1}{2}}{\binom{20}{3}} \qquad i = 3, \ldots, 20 \tag{1.1}$$

Equation (1.1) follows because the number of selections that result in the event $\{X = i\}$ is just the number of selections that result in the ball numbered i and two of the balls numbered 1 through $i - 1$ being chosen. Because there are clearly $\binom{1}{1}\binom{i-1}{2}$ such selections, we obtain the probabilities expressed in Equation (1.1), from which we see that

$$P\{X = 20\} = \frac{\binom{19}{2}}{\binom{20}{3}} = \frac{3}{20} = .150$$

$$P\{X = 19\} = \frac{\binom{18}{2}}{\binom{20}{3}} = \frac{51}{380} \approx .134$$

$$P\{X = 18\} = \frac{\binom{17}{2}}{\binom{20}{3}} = \frac{34}{285} \approx .119$$

$$P\{X = 17\} = \frac{\binom{16}{2}}{\binom{20}{3}} = \frac{2}{19} \approx .105$$

Hence, since the event $\{X \geq 17\}$ is the union of the disjoint events $\{X = i\}$, $i = 17, 18, 19, 20$, it follows that the probability of our winning the bet is given by

$$P\{X \geq 17\} \approx .105 + .119 + .134 + .150 = .508$$ ■

EXAMPLE 1c

Independent trials consisting of the flipping of a coin having probability p of coming up heads are continually performed until either a head occurs or a total of n flips is made. If we let X denote the number of times the coin is flipped, then X is a random variable taking on one of the values $1, 2, 3, \ldots, n$ with respective probabilities

$$P\{X = 1\} = P\{H\} = p$$

$$P\{X = 2\} = P\{(T, H)\} = (1 - p)p$$

$$P\{X = 3\} = P\{(T, T, H)\} = (1 - p)^2 p$$

$$\vdots$$

$$P\{X = n - 1\} = P\{(\underbrace{T, T, \ldots, T}_{n-2}, H)\} = (1 - p)^{n-2} p$$

$$P\{X = n\} = P\{(\underbrace{T, T, \ldots, T}_{n-1}, T), (\underbrace{T, T, \ldots, T}_{n-1}, H)\} = (1 - p)^{n-1}$$

As a check, note that

$$\begin{aligned} P\left(\bigcup_{i=1}^{n} \{X = i\}\right) &= \sum_{i=1}^{n} P\{X = i\} \\ &= \sum_{i=1}^{n-1} p(1 - p)^{i-1} + (1 - p)^{n-1} \\ &= p\left[\frac{1 - (1 - p)^{n-1}}{1 - (1 - p)}\right] + (1 - p)^{n-1} \\ &= 1 - (1 - p)^{n-1} + (1 - p)^{n-1} \\ &= 1 \end{aligned}$$

■

EXAMPLE 1d

Three balls are randomly chosen from an urn containing 3 white, 3 red, and 5 black balls. Suppose that we win \$1 for each white ball selected and lose \$1 for each red ball

selected. If we let X denote our total winnings from the experiment, then X is a random variable taking on the possible values $0, \pm 1, \pm 2, \pm 3$ with respective probabilities

$$P\{X = 0\} = \frac{\binom{5}{3} + \binom{3}{1}\binom{3}{1}\binom{5}{1}}{\binom{11}{3}} = \frac{55}{165}$$

$$P\{X = 1\} = P\{X = -1\} = \frac{\binom{3}{1}\binom{5}{2} + \binom{3}{2}\binom{3}{1}}{\binom{11}{3}} = \frac{39}{165}$$

$$P\{X = 2\} = P\{X = -2\} = \frac{\binom{3}{2}\binom{5}{1}}{\binom{11}{3}} = \frac{15}{165}$$

$$P\{X = 3\} = P\{X = -3\} = \frac{\binom{3}{3}}{\binom{11}{3}} = \frac{1}{165}$$

These probabilities are obtained, for instance, by noting that in order for X to equal 0, either all 3 balls selected must be black or 1 ball of each color must be selected. Similarly, the event $\{X = 1\}$ occurs either if 1 white and 2 black balls are selected or if 2 white and 1 red is selected. As a check, we note that

$$\sum_{i=0}^{3} P\{X = i\} + \sum_{i=1}^{3} P\{X = -i\} = \frac{55 + 39 + 15 + 1 + 39 + 15 + 1}{165} = 1$$

The probability that we win money is given by

$$\sum_{i=1}^{3} P\{X = i\} = \frac{55}{165} = \frac{1}{3}$$

■

EXAMPLE 1e

Suppose that there are N distinct types of coupons and that each time one obtains a coupon, it is, independently of previous selections, equally likely to be any one of the N types. One random variable of interest is T, the number of coupons that needs to be collected until one obtains a complete set of at least one of each type. Rather than derive $P\{T = n\}$ directly, let us start by considering the probability that T is greater than n. To do so, fix n and define the events $A_1, A_2, \ldots, A_N$ as follows: A_j is the event

that no type j coupon is contained among the first n coupons collected, $j = 1, \ldots, N$. Hence,

$$\begin{aligned} P\{T > n\} &= P\left(\bigcup_{j=1}^{N} A_j\right) \\ &= \sum_j P(A_j) - \sum_{j_1<j_2}\sum P(A_{j_1}A_{j_2}) + \cdots \\ &\quad + (-1)^{k+1}\sum_{j_1<j_2<\cdots<j_k}\sum\cdots\sum P(A_{j_1}A_{j_2}\cdots A_{j_k})\cdots \\ &\quad + (-1)^{N+1}P(A_1A_2\cdots A_N) \end{aligned}$$

Now, A_j will occur if each of the n coupons collected is not of type j. Since each of the coupons will not be of type j with probability $(N-1)/N$, we have, by the assumed independence of the types of successive coupons,

$$P(A_j) = \left(\frac{N-1}{N}\right)^n$$

Also, the event $A_{j_1}A_{j_2}$ will occur if none of the first n coupons collected is of either type j_1 or type j_2. Thus, again using independence, we see that

$$P(A_{j_1}A_{j_2}) = \left(\frac{N-2}{N}\right)^n$$

The same reasoning gives

$$P(A_{j_1}A_{j_2}\cdots A_{j_k}) = \left(\frac{N-k}{N}\right)^n$$

and we see that, for $n > 0$,

$$\begin{aligned} P\{T > n\} &= N\left(\frac{N-1}{N}\right)^n - \binom{N}{2}\left(\frac{N-2}{N}\right)^n + \binom{N}{3}\left(\frac{N-3}{N}\right)^n - \cdots \\ &\quad + (-1)^N\binom{N}{N-1}\left(\frac{1}{N}\right)^n \\ &= \sum_{i=1}^{N-1}\binom{N}{i}\left(\frac{N-i}{N}\right)^n(-1)^{i+1} \end{aligned} \tag{1.2}$$

The probability that T equals n can now be obtained from the preceding formula by the use of

$$P\{T > n-1\} = P\{T = n\} + P\{T > n\}$$

or, equivalently,

$$P\{T = n\} = P\{T > n-1\} - P\{T > n\}$$

Another random variable of interest is the number of distinct types of coupons that are contained in the first n selections—call this random variable D_n. To compute

$P\{D_n = k\}$, let us start by fixing attention on a particular set of k distinct types, and let us then determine the probability that this set constitutes the set of distinct types obtained in the first n selections. Now, in order for this to be the situation, it is necessary and sufficient that, of the first n coupons obtained,

A: each is one of these k types.
B: each of these k types is represented.

Now, each coupon selected will be one of the k types with probability k/N, so the probability that A will be valid is $(k/N)^n$. Also, given that a coupon is of one of the k types under consideration, it is easy to see that it is equally likely to be of any one of these k types. Hence, the conditional probability of B given that A occurs is the same as the probability that a set of n coupons, each equally likely to be any of k possible types, contains a complete set of all k types. But this is just the probability that the number needed to amass a complete set, when choosing among k types, is less than or equal to n and is thus obtainable from Equation (1.2) with k replacing N. Thus, we have

$$P(A) = \left(\frac{k}{N}\right)^n$$

$$P(B|A) = 1 - \sum_{i=1}^{k-1} \binom{k}{i} \left(\frac{k-i}{k}\right)^n (-1)^{i+1}$$

Finally, as there are $\binom{N}{k}$ possible choices for the set of k types, we arrive at

$$\begin{aligned} P\{D_n = k\} &= \binom{N}{k} P(AB) \\ &= \binom{N}{k} \left(\frac{k}{N}\right)^n \left[1 - \sum_{i=1}^{k-1} \binom{k}{i} \left(\frac{k-i}{k}\right)^n (-1)^{i+1}\right] \end{aligned}$$

Remark. Since one must collect at least N coupons to obtain a compete set, it follows that $P\{T > n\} = 1$ if $n < N$. Therefore, from Equation (1.2), we obtain the interesting combinatorial identity that, for integers $1 \le n < N$,

$$\sum_{i=1}^{N-1} \binom{N}{i} \left(\frac{N-i}{N}\right)^n (-1)^{i+1} = 1$$

which can be written as

$$\sum_{i=0}^{N-1} \binom{N}{i} \left(\frac{N-i}{N}\right)^n (-1)^{i+1} = 0$$

or, upon multiplying by $(-1)^N N^n$ and letting $j = N - i$,

$$\sum_{j=1}^{N} \binom{N}{j} j^n (-1)^{j-1} = 0 \qquad 1 \le n < N$$

■

For a random variable X, the function F defined by

$$F(x) = P\{X \leq x\} \qquad -\infty < x < \infty$$

is called the *cumulative distribution function*, or, more simply, the *distribution function*, of X. Thus, the distribution function specifies, for all real values x, the probability that the random variable is less than or equal to x.

Now, suppose that $a \leq b$. Then, because the event $\{X \leq a\}$ is contained in the event $\{X \leq b\}$, it follows that $F(a)$, the probability of the former, is less than or equal to $F(b)$, the probability of the latter. In other words, $F(x)$ is a nondecreasing function of x. Other general properties of the distribution function are given in Section 4.10.

4.2 DISCRETE RANDOM VARIABLES

A random variable that can take on at most a countable number of possible values is said to be discrete. For a discrete random variable X, we define the *probability mass function* $p(a)$ of X by

$$p(a) = P\{X = a\}$$

The probability mass function $p(a)$ is positive for at most a countable number of values of a. That is, if X must assume one of the values $x_1, x_2, \ldots$, then

$$\begin{aligned} p(x_i) &\geq 0 \quad \text{for } i = 1, 2, \ldots \\ p(x) &= 0 \quad \text{for all other values of } x \end{aligned}$$

Since X must take on one of the values x_i, we have

$$\sum_{i=1}^{\infty} p(x_i) = 1$$

It is often instructive to present the probability mass function in a graphical format by plotting $p(x_i)$ on the y-axis against x_i on the x-axis. For instance, if the probability mass function of X is

$$p(0) = \frac{1}{4} \qquad p(1) = \frac{1}{2} \qquad p(2) = \frac{1}{4}$$

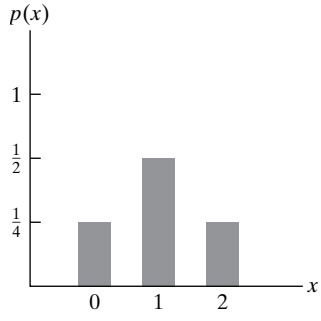


FIGURE 4.1

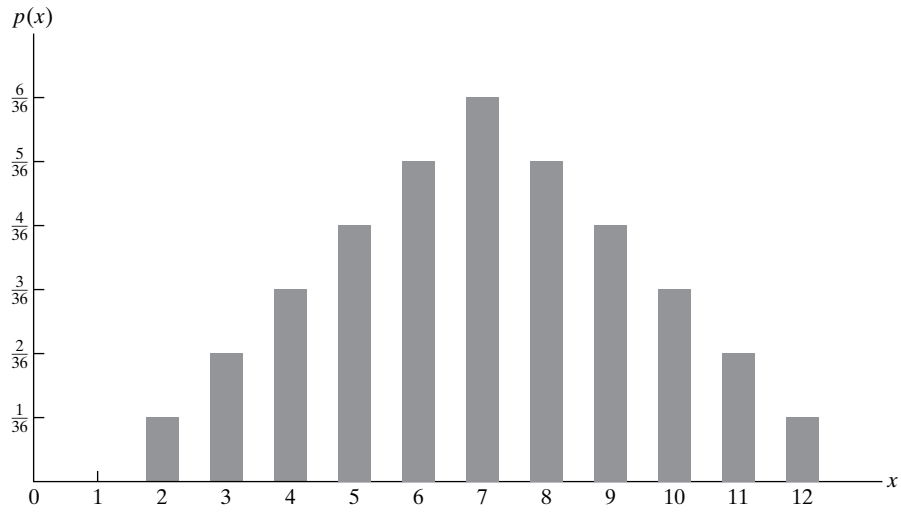


FIGURE 4.2

we can represent this function graphically as shown in Figure 4.1. Similarly, a graph of the probability mass function of the random variable representing the sum when two dice are rolled looks like Figure 4.2.

EXAMPLE 2a

The probability mass function of a random variable X is given by $p(i) = c\lambda^i/i!$, $i = 0, 1, 2, \ldots$, where λ is some positive value. Find (a) $P\{X = 0\}$ and (b) $P\{X > 2\}$.

Solution. Since $\sum_{i=0}^{\infty} p(i) = 1$, we have

$$c\sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = 1$$

which, because $e^x = \sum_{i=0}^{\infty} x^i/i!$, implies that

$$ce^{\lambda} = 1 \quad \text{or} \quad c = e^{-\lambda}$$

Hence,

(a) $P\{X = 0\} = e^{-\lambda}\lambda^0/0! = e^{-\lambda}$

(b) $P\{X > 2\} = 1 - P\{X \leq 2\} = 1 - P\{X = 0\} - P\{X = 1\} - P\{X = 2\}$

$$= 1 - e^{-\lambda} - \lambda e^{-\lambda} - \frac{\lambda^2 e^{-\lambda}}{2}$$

■

The cumulative distribution function F can be expressed in terms of $p(a)$ by

$$F(a) = \sum_{\text{all } x \leq a} p(x)$$

If X is a discrete random variable whose possible values are $x_1, x_2, x_3, \ldots$, where $x_1 < x_2 < x_3 < \cdots$, then the distribution function F of X is a step function. That is,

the value of F is constant in the intervals $[x_{i-1}, x_i)$ and then takes a step (or jump) of size $p(x_i)$ at x_i. For instance, if X has a probability mass function given by

$$p(1) = \frac{1}{4} \qquad p(2) = \frac{1}{2} \qquad p(3) = \frac{1}{8} \qquad p(4) = \frac{1}{8}$$

then its cumulative distribution function is

$$F(a) = \begin{cases} 0 & a < 1 \\ \frac{1}{4} & 1 \le a < 2 \\ \frac{3}{4} & 2 \le a < 3 \\ \frac{7}{8} & 3 \le a < 4 \\ 1 & 4 \le a \end{cases}$$

This function is depicted graphically in Figure 4.3.

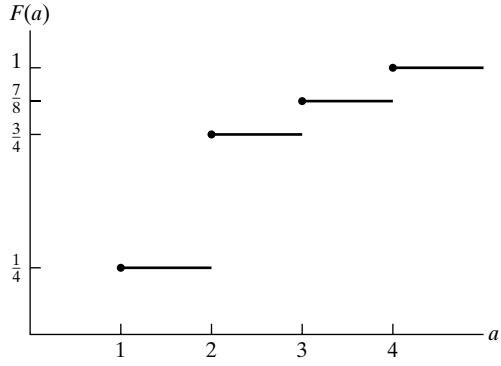


FIGURE 4.3

Note that the size of the step at any of the values 1, 2, 3, and 4 is equal to the probability that X assumes that particular value.

4.3 EXPECTED VALUE

One of the most important concepts in probability theory is that of the expectation of a random variable. If X is a discrete random variable having a probability mass function $p(x)$, then the *expectation*, or the *expected value*, of X, denoted by $E[X]$, is defined by

$$E[X] = \sum_{x:p(x)>0} xp(x)$$

In words, the expected value of X is a weighted average of the possible values that X can take on, each value being weighted by the probability that X assumes it. For instance, on the one hand, if the probability mass function of X is given by

$$p(0) = \frac{1}{2} = p(1)$$

then

$$E[X] = 0\left(\frac{1}{2}\right) + 1\left(\frac{1}{2}\right) = \frac{1}{2}$$

is just the ordinary average of the two possible values, 0 and 1, that X can assume. On the other hand, if

$$p(0) = \frac{1}{3} \qquad p(1) = \frac{2}{3}$$

then

$$E[X] = 0\left(\frac{1}{3}\right) + 1\left(\frac{2}{3}\right) = \frac{2}{3}$$

is a weighted average of the two possible values 0 and 1, where the value 1 is given twice as much weight as the value 0, since $p(1) = 2p(0)$.

Another motivation of the definition of expectation is provided by the frequency interpretation of probabilities. This interpretation (partially justified by the strong law of large numbers, to be presented in Chapter 8) assumes that if an infinite sequence of independent replications of an experiment is performed, then, for any event E, the proportion of time that E occurs will be $P(E)$. Now, consider a random variable X that must take on one of the values $x_1, x_2, \ldots x_n$ with respective probabilities $p(x_1), p(x_2), \ldots, p(x_n)$, and think of X as representing our winnings in a single game of chance. That is, with probability $p(x_i)$ we shall win x_i units $i = 1, 2, \ldots, n$. By the frequency interpretation, if we play this game continually, then the proportion of time that we win x_i will be $p(x_i)$. Since this is true for all $i, i = 1, 2, \ldots, n$, it follows that our average winnings per game will be

$$\sum_{i=1}^{n} x_i p(x_i) = E[X]$$

EXAMPLE 3a

Find $E[X]$, where X is the outcome when we roll a fair die.

Solution. Since $p(1) = p(2) = p(3) = p(4) = p(5) = p(6) = \frac{1}{6}$, we obtain

$$E[X] = 1\left(\frac{1}{6}\right) + 2\left(\frac{1}{6}\right) + 3\left(\frac{1}{6}\right) + 4\left(\frac{1}{6}\right) + 5\left(\frac{1}{6}\right) + 6\left(\frac{1}{6}\right) = \frac{7}{2}$$

■

EXAMPLE 3b

We say that I is an indicator variable for the event A if

$$I = \begin{cases} 1 & \text{if } A \text{ occurs} \\ 0 & \text{if } A^c \text{ occurs} \end{cases}$$

Find $E[I]$.

Solution. Since $p(1) = P(A), p(0) = 1 - P(A)$, we have

$$E[I] = P(A)$$

That is, the expected value of the indicator variable for the event A is equal to the probability that A occurs. ■

EXAMPLE 3c

A contestant on a quiz show is presented with two questions, questions 1 and 2, which he is to attempt to answer in some order he chooses. If he decides to try question i first, then he will be allowed to go on to question j, $j \neq i$, only if his answer to question i is correct. If his initial answer is incorrect, he is not allowed to answer the other question. The contestant is to receive V_i dollars if he answers question i correctly, $i = 1, 2$. For instance, he will receive $V_1 + V_2$ dollars if he answers both questions correctly. If the probability that he knows the answer to question i is $P_i, i = 1, 2$, which question should he attempt to answer first so as to maximize his expected winnings? Assume that the events $E_i, i = 1, 2$, that he knows the answer to question i are independent events.

Solution. On the one hand, if he attempts to answer question 1 first, then he will win

$$\begin{array}{ll} 0 & \text{with probability } 1 - P_1 \\ V_1 & \text{with probability } P_1(1 - P_2) \\ V_1 + V_2 & \text{with probability } P_1P_2 \end{array}$$

Hence, his expected winnings in this case will be

$$V_1P_1(1 - P_2) + (V_1 + V_2)P_1P_2$$

On the other hand, if he attempts to answer question 2 first, his expected winnings will be

$$V_2P_2(1 - P_1) + (V_1 + V_2)P_1P_2$$

Therefore, it is better to try question 1 first if

$$V_1P_1(1 - P_2) \geq V_2P_2(1 - P_1)$$

or, equivalently, if

$$\frac{V_1P_1}{1 - P_1} \geq \frac{V_2P_2}{1 - P_2}$$

For example, if he is 60 percent certain of answering question 1, worth \$200, correctly and he is 80 percent certain of answering question 2, worth \$100, correctly, then he should attempt to answer question 2 first because

$$400 = \frac{(100)(.8)}{.2} > \frac{(200)(.6)}{.4} = 300$$ ■

EXAMPLE 3d

A school class of 120 students is driven in 3 buses to a symphonic performance. There are 36 students in one of the buses, 40 in another, and 44 in the third bus. When the buses arrive, one of the 120 students is randomly chosen. Let X denote the number of students on the bus of that randomly chosen student, and find $E[X]$.

Solution. Since the randomly chosen student is equally likely to be any of the 120 students, it follows that

$$P\{X = 36\} = \frac{36}{120} \quad P\{X = 40\} = \frac{40}{120} \quad P\{X = 44\} = \frac{44}{120}$$

Hence,

$$E[X] = 36\left(\frac{3}{10}\right) + 40\left(\frac{1}{3}\right) + 44\left(\frac{11}{30}\right) = \frac{1208}{30} = 40.2667$$

However, the average number of students on a bus is $120/3 = 40$, showing that the expected number of students on the bus of a randomly chosen student is larger than the average number of students on a bus. This is a general phenomenon, and it occurs because the more students there are on a bus, the more likely it is that a randomly chosen student would have been on that bus. As a result, buses with many students are given more weight than those with fewer students. (See Self-Test Problem 4.) ■

Remark. The probability concept of expectation is analogous to the physical concept of the *center of gravity* of a distribution of mass. Consider a discrete random variable X having probability mass function $p(x_i), i \geq 1$. If we now imagine a weightless rod in which weights with mass $p(x_i), i \geq 1$, are located at the points $x_i, i \geq 1$ (see Figure 4.4), then the point at which the rod would be in balance is known as the center of gravity. For those readers acquainted with elementary statics, it is now a simple matter to show that this point is at $E[X]$.† ■

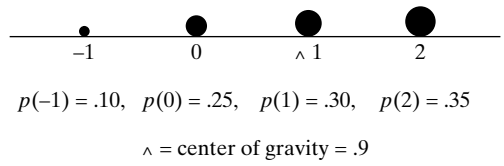


FIGURE 4.4

4.4 EXPECTATION OF A FUNCTION OF A RANDOM VARIABLE

Suppose that we are given a discrete random variable along with its probability mass function and that we want to compute the expected value of some function of X, say, $g(X)$. How can we accomplish this? One way is as follows: Since $g(X)$ is itself a discrete random variable, it has a probability mass function, which can be determined from the probability mass function of X. Once we have determined the probability mass function of $g(X)$, we can compute $E[g(X)]$ by using the definition of expected value.

EXAMPLE 4a

Let X denote a random variable that takes on any of the values -1, 0, and 1 with respective probabilities

$$P\{X = -1\} = .2 \quad P\{X = 0\} = .5 \quad P\{X = 1\} = .3$$

Compute $E[X^2]$.

Solution. Let $Y = X^2$. Then the probability mass function of Y is given by

$$P\{Y = 1\} = P\{X = -1\} + P\{X = 1\} = .5$$
$$P\{Y = 0\} = P\{X = 0\} = .5$$

Hence,

$$E[X^2] = E[Y] = 1(.5) + 0(.5) = .5$$

†To prove this, we must show that the sum of the torques tending to turn the point around $E[X]$ is equal to 0. That is, we must show that $0 = \sum_i (x_i - E[X])p(x_i)$, which is immediate.

Note that

$$.5 = E[X^2] \neq (E[X])^2 = .01$$ ■

Although the preceding procedure will always enable us to compute the expected value of any function of X from a knowledge of the probability mass function of X, there is another way of thinking about $E[g(X)]$: Since $g(X)$ will equal $g(x)$ whenever X is equal to x, it seems reasonable that $E[g(X)]$ should just be a weighted average of the values $g(x)$, with $g(x)$ being weighted by the probability that X is equal to x. That is, the following result is quite intuitive:

Proposition 4.1.
If X is a discrete random variable that takes on one of the values $x_i, i \geq 1$, with respective probabilities $p(x_i)$, then, for any real-valued function g,

$$E[g(X)] = \sum_i g(x_i)p(x_i)$$

Before proving this proposition, let us check that it is in accord with the results of Example 4a. Applying it to that example yields

$$\begin{aligned} E\{X^2\} &= (-1)^2(.2) + 0^2(.5) + 1^2(.3) \\ &= 1(.2 + .3) + 0(.5) \\ &= .5 \end{aligned}$$

which is in agreement with the result given in Example 4a.

Proof of Proposition 4.1: The proof of Proposition 4.1 proceeds, as in the preceding verification, by grouping together all the terms in $\sum_i g(x_i)p(x_i)$ having the same value of $g(x_i)$. Specifically, suppose that $y_j, j \geq 1$, represent the different values of $g(x_i), i \geq 1$. Then, grouping all the $g(x_i)$ having the same value gives

$$\begin{aligned} \sum_i g(x_i)p(x_i) &= \sum_j \sum_{i:g(x_i)=y_j} g(x_i)p(x_i) \\ &= \sum_j \sum_{i:g(x_i)=y_j} y_j p(x_i) \\ &= \sum_j y_j \sum_{i:g(x_i)=y_j} p(x_i) \\ &= \sum_j y_j P\{g(X) = y_j\} \\ &= E[g(X)] \end{aligned}$$ □

EXAMPLE 4b

A product that is sold seasonally yields a net profit of b dollars for each unit sold and a net loss of ℓ dollars for each unit left unsold when the season ends. The number of units of the product that are ordered at a specific department store during any season is a random variable having probability mass function $p(i), i \geq 0$. If the store must stock this product in advance, determine the number of units the store should stock so as to maximize its expected profit.

Solution. Let X denote the number of units ordered. If s units are stocked, then the profit—call it $P(s)$—can be expressed as

$$\begin{aligned} P(s) &= bX - (s - X)\ell \quad \text{if } X \le s \\ &= sb \qquad\qquad\qquad\quad \text{if } X > s \end{aligned}$$

Hence, the expected profit equals

$$\begin{aligned} E[P(s)] &= \sum_{i=0}^{s} [bi - (s - i)\ell]p(i) + \sum_{i=s+1}^{\infty} sbp(i) \\ &= (b + \ell)\sum_{i=0}^{s} ip(i) - s\ell\sum_{i=0}^{s} p(i) + sb\left[1 - \sum_{i=0}^{s} p(i)\right] \\ &= (b + \ell)\sum_{i=0}^{s} ip(i) - (b + \ell)s\sum_{i=0}^{s} p(i) + sb \\ &= sb + (b + \ell)\sum_{i=0}^{s} (i - s)p(i) \end{aligned}$$

To determine the optimum value of s, let us investigate what happens to the profit when we increase s by 1 unit. By substitution, we see that the expected profit in this case is given by

$$\begin{aligned} E[P(s + 1)] &= b(s + 1) + (b + \ell)\sum_{i=0}^{s+1} (i - s - 1)p(i) \\ &= b(s + 1) + (b + \ell)\sum_{i=0}^{s} (i - s - 1)p(i) \end{aligned}$$

Therefore,

$$E[P(s + 1)] - E[P(s)] = b - (b + \ell)\sum_{i=0}^{s} p(i)$$

Thus, stocking $s + 1$ units will be better than stocking s units whenever

$$\sum_{i=0}^{s} p(i) < \frac{b}{b + \ell} \tag{4.1}$$

Because the left-hand side of Equation (4.1) is increasing in s while the right-hand side is constant, the inequality will be satisfied for all values of $s \le s^*$, where s^* is the largest value of s satisfying Equation (4.1). Since

$$E[P(0)] < \cdots < E[P(s^*)] < E[P(s^* + 1)] > E[P(s^* + 2)] > \cdots$$

it follows that stocking $s^* + 1$ items will lead to a maximum expected profit. ■

EXAMPLE 4c Utility

Suppose that you must choose one of two possible actions, each of which can result in any of n consequences, denoted as $C_1, \ldots, C_n$. Suppose that if the first action is

chosen, then consequence C_i will result with probability $p_i, i = 1, \ldots, n$, whereas if the second action is chosen, then consequence C_i will result with probability q_i, $i = 1, \ldots, n$, where $\sum_{i=1}^{n} p_i = \sum_{i=1}^{n} q_i = 1$. The following approach can be used to determine which action to choose: Start by assigning numerical values to the different consequences in the following manner: First, identify the least and the most desirable consequences—call them c and C, respectively; give consequence c the value 0 and give C the value 1. Now consider any of the other $n - 2$ consequences, say, C_i. To value this consequence, imagine that you are given the choice between either receiving C_i or taking part in a random experiment that either earns you consequence C with probability u or consequence c with probability $1 - u$. Clearly, your choice will depend on the value of u. On the one hand, if $u = 1$, then the experiment is certain to result in consequence C, and since C is the most desirable consequence, you will prefer participating in the experiment to receiving C_i. On the other hand, if $u = 0$, then the experiment will result in the least desirable consequence—namely, c—so in this case you will prefer the consequence C_i to participating in the experiment. Now, as u decreases from 1 to 0, it seems reasonable that your choice will at some point switch from participating in the experiment to the certain return of C_i, and at that critical switch point you will be indifferent between the two alternatives. Take that indifference probability u as the value of the consequence C_i. In other words, the value of C_i is that probability u such that you are indifferent between either receiving the consequence C_i or taking part in an experiment that returns consequence C with probability u or consequence c with probability $1 - u$. We call this indifference probability the *utility* of the consequence C_i, and we designate it as $u(C_i)$.

To determine which action is superior, we need to evaluate each one. Consider the first action, which results in consequence C_i with probability $p_i, i = 1, \ldots, n$. We can think of the result of this action as being determined by a two-stage experiment. In the first stage, one of the values $1, \ldots, n$ is chosen according to the probabilities $p_1, \ldots, p_n$; if value i is chosen, you receive consequence C_i. However, since C_i is equivalent to obtaining consequence C with probability $u(C_i)$ or consequence c with probability $1 - u(C_i)$, it follows that the result of the two-stage experiment is equivalent to an experiment in which either consequence C or consequence c is obtained, with C being obtained with probability

$$\sum_{i=1}^{n} p_i u(C_i)$$

Similarly, the result of choosing the second action is equivalent to taking part in an experiment in which either consequence C or consequence c is obtained, with C being obtained with probability

$$\sum_{i=1}^{n} q_i u(C_i)$$

Since C is preferable to c, it follows that the first action is preferable to the second action if

$$\sum_{i=1}^{n} p_i u(C_i) > \sum_{i=1}^{n} q_i u(C_i)$$

In other words, the worth of an action can be measured by the expected value of the utility of its consequence, and the action with the largest expected utility is the most preferable. ■

A simple logical consequence of Proposition 4.1 is Corollary 4.1.

Corollary 4.1. If a and b are constants, then

$$E[aX + b] = aE[X] + b$$

Proof.

$$\begin{aligned} E[aX + b] &= \sum_{x:p(x)>0} (ax + b)p(x) \\ &= a \sum_{x:p(x)>0} xp(x) + b \sum_{x:p(x)>0} p(x) \\ &= aE[X] + b \end{aligned}$$

□

The expected value of a random variable X, $E[X]$, is also referred to as the *mean* or the *first moment of X*. The quantity $E[X^n], n \geq 1$, is called the *nth moment of X*. By Proposition 4.1, we note that

$$E[X^n] = \sum_{x:p(x)>0} x^n p(x)$$

4.5 VARIANCE

Given a random variable X along with its distribution function F, it would be extremely useful if we were able to summarize the essential properties of F by certain suitably defined measures. One such measure would be $E[X]$, the expected value of X. However, although $E[X]$ yields the weighted average of the possible values of X, it does not tell us anything about the variation, or spread, of these values. For instance, although random variables W, Y, and Z having probability mass functions determined by

$$W = 0 \quad \text{with probability 1}$$

$$Y = \begin{cases} -1 & \text{with probability } \frac{1}{2} \\ +1 & \text{with probability } \frac{1}{2} \end{cases}$$

$$Z = \begin{cases} -100 & \text{with probability } \frac{1}{2} \\ +100 & \text{with probability } \frac{1}{2} \end{cases}$$

all have the same expectation—namely, 0—there is a much greater spread in the possible values of Y than in those of W (which is a constant) and in the possible values of Z than in those of Y.

Because we expect X to take on values around its mean $E[X]$, it would appear that a reasonable way of measuring the possible variation of X would be to look at how far apart X would be from its mean, on the average. One possible way to measure this variation would be to consider the quantity $E[|X - \mu|]$, where $\mu = E[X]$. However, it turns out to be mathematically inconvenient to deal with this quantity, so a more tractable quantity is usually considered—namely, the expectation of the square of the difference between X and its mean. We thus have the following definition.

Definition

If X is a random variable with mean μ, then the variance of X, denoted by $\text{Var}(X)$, is defined by

$$\text{Var}(X) = E[(X - \mu)^2]$$

An alternative formula for $\text{Var}(X)$ is derived as follows:

$$\begin{aligned}
\text{Var}(X) &= E[(X - \mu)^2] \\
&= \sum_x (x - \mu)^2 p(x) \\
&= \sum_x (x^2 - 2\mu x + \mu^2) p(x) \\
&= \sum_x x^2 p(x) - 2\mu \sum_x x p(x) + \mu^2 \sum_x p(x) \\
&= E[X^2] - 2\mu^2 + \mu^2 \\
&= E[X^2] - \mu^2
\end{aligned}$$

That is,

$$\boxed{\text{Var}(X) = E[X^2] - (E[X])^2}$$

In words, the variance of X is equal to the expected value of X^2 minus the square of its expected value. In practice, this formula frequently offers the easiest way to compute $\text{Var}(X)$.

EXAMPLE 5a

Calculate $\text{Var}(X)$ if X represents the outcome when a fair die is rolled.

Solution. It was shown in Example 3a that $E[X] = \frac{7}{2}$. Also,

$$\begin{aligned}
E[X^2] &= 1^2\left(\frac{1}{6}\right) + 2^2\left(\frac{1}{6}\right) + 3^2\left(\frac{1}{6}\right) + 4^2\left(\frac{1}{6}\right) + 5^2\left(\frac{1}{6}\right) + 6^2\left(\frac{1}{6}\right) \\
&= \left(\frac{1}{6}\right)(91)
\end{aligned}$$

Hence,

$$\text{Var}(X) = \frac{91}{6} - \left(\frac{7}{2}\right)^2 = \frac{35}{12}$$

■

A useful identity is that, for any constants a and b,

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

To prove this equality, let $\mu = E[X]$ and note from Corollary 4.1 that $E[aX + b] = a\mu + b$. Therefore,

$$\begin{aligned}\text{Var}(aX + b) &= E[(aX + b - a\mu - b)^2] \\ &= E[a^2(X - \mu)^2] \\ &= a^2E[(X - \mu)^2] \\ &= a^2\text{Var}(X)\end{aligned}$$

Remarks. (a) Analogous to the means being the center of gravity of a distribution of mass, the variance represents, in the terminology of mechanics, the moment of inertia.

(b) The square root of the $\text{Var}(X)$ is called the *standard deviation* of X, and we denote it by $\text{SD}(X)$. That is,

$$\text{SD}(X) = \sqrt{\text{Var}(X)}$$

Discrete random variables are often classified according to their probability mass functions. In the next few sections, we consider some of the more common types.

4.6 THE BERNOULLI AND BINOMIAL RANDOM VARIABLES

Suppose that a trial, or an experiment, whose outcome can be classified as either a *success* or a *failure* is performed. If we let $X = 1$ when the outcome is a success and $X = 0$ when it is a failure, then the probability mass function of X is given by

$$\begin{aligned}p(0) &= P\{X = 0\} = 1 - p \\ p(1) &= P\{X = 1\} = p\end{aligned} \tag{6.1}$$

where $p, 0 \le p \le 1$, is the probability that the trial is a success.

A random variable X is said to be a *Bernoulli random variable* (after the Swiss mathematician James Bernoulli) if its probability mass function is given by Equations (6.1) for some $p \in (0, 1)$.

Suppose now that n independent trials, each of which results in a success with probability p and in a failure with probability $1 - p$, are to be performed. If X represents the number of successes that occur in the n trials, then X is said to be a *binomial random variable* with parameters (n, p). Thus, a Bernoulli random variable is just a binomial random variable with parameters $(1, p)$.

The probability mass function of a binomial random variable having parameters (n, p) is given by

$$p(i) = \binom{n}{i} p^i(1 - p)^{n-i} \qquad i = 0, 1, \ldots, n \tag{6.2}$$

The validity of Equation (6.2) may be verified by first noting that the probability of any particular sequence of n outcomes containing i successes and $n - i$ failures is, by the assumed independence of trials, $p^i(1 - p)^{n-i}$. Equation (6.2) then follows, since there are $\binom{n}{i}$ different sequences of the n outcomes leading to i successes and $n - i$ failures. This perhaps can most easily be seen by noting that there are $\binom{n}{i}$ different choices of the i trials that result in successes. For instance, if $n = 4, i = 2$, then there are $\binom{4}{2} = 6$ ways in which the four trials can result in two successes,

namely, any of the outcomes (s, s, f, f), (s, f, s, f), (s, f, f, s), (f, s, s, f), (f, s, f, s), and (f, f, s, s), where the outcome (s, s, f, f) means, for instance, that the first two trials are successes and the last two failures. Since each of these outcomes has probability $p^2(1 - p)^2$ of occurring, the desired probability of two successes in the four trials is $\binom{4}{2} p^2(1 - p)^2$.

Note that, by the binomial theorem, the probabilities sum to 1; that is,

$$\sum_{i=0}^{\infty} p(i) = \sum_{i=0}^{n} \binom{n}{i} p^i(1 - p)^{n-i} = [p + (1 - p)]^n = 1$$

EXAMPLE 6a

Five fair coins are flipped. If the outcomes are assumed independent, find the probability mass function of the number of heads obtained.

Solution. If we let X equal the number of heads (successes) that appear, then X is a binomial random variable with parameters $\left(n = 5, p = \frac{1}{2}\right)$. Hence, by Equation (6.2),

$$P\{X = 0\} = \binom{5}{0} \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^5 = \frac{1}{32}$$

$$P\{X = 1\} = \binom{5}{1} \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^4 = \frac{5}{32}$$

$$P\{X = 2\} = \binom{5}{2} \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^3 = \frac{10}{32}$$

$$P\{X = 3\} = \binom{5}{3} \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^2 = \frac{10}{32}$$

$$P\{X = 4\} = \binom{5}{4} \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^1 = \frac{5}{32}$$

$$P\{X = 5\} = \binom{5}{5} \left(\frac{1}{2}\right)^5 \left(\frac{1}{2}\right)^0 = \frac{1}{32}$$

■

EXAMPLE 6b

It is known that screws produced by a certain company will be defective with probability .01, independently of each other. The company sells the screws in packages of 10 and offers a money-back guarantee that at most 1 of the 10 screws is defective. What proportion of packages sold must the company replace?

Solution. If X is the number of defective screws in a package, then X is a binomial random variable with parameters (10, .01). Hence, the probability that a package will have to be replaced is

$$1 - P\{X = 0\} - P\{X = 1\} = 1 - \binom{10}{0} (.01)^0 (.99)^{10} - \binom{10}{1} (.01)^1 (.99)^9$$
$$\approx .004$$

Thus, only .4 percent of the packages will have to be replaced. ■

EXAMPLE 6c

The following gambling game, known as the wheel of fortune (or chuck-a-luck), is quite popular at many carnivals and gambling casinos: A player bets on one of the numbers 1 through 6. Three dice are then rolled, and if the number bet by the player appears i times, $i = 1, 2, 3$, then the player wins i units; if the number bet by the player does not appear on any of the dice, then the player loses 1 unit. Is this game fair to the player? (Actually, the game is played by spinning a wheel that comes to rest on a slot labeled by three of the numbers 1 through 6, but this variant is mathematically equivalent to the dice version.)

Solution. If we assume that the dice are fair and act independently of each other, then the number of times that the number bet appears is a binomial random variable with parameters $\left(3, \frac{1}{6}\right)$. Hence, letting X denote the player's winnings in the game, we have

$$P\{X = -1\} = \binom{3}{0}\left(\frac{1}{6}\right)^0\left(\frac{5}{6}\right)^3 = \frac{125}{216}$$

$$P\{X = 1\} = \binom{3}{1}\left(\frac{1}{6}\right)^1\left(\frac{5}{6}\right)^2 = \frac{75}{216}$$

$$P\{X = 2\} = \binom{3}{2}\left(\frac{1}{6}\right)^2\left(\frac{5}{6}\right)^1 = \frac{15}{216}$$

$$P\{X = 3\} = \binom{3}{3}\left(\frac{1}{6}\right)^3\left(\frac{5}{6}\right)^0 = \frac{1}{216}$$

In order to determine whether or not this is a fair game for the player, let us calculate $E[X]$. From the preceding probabilities, we obtain

$$E[X] = \frac{-125 + 75 + 30 + 3}{216}$$
$$= \frac{-17}{216}$$

Hence, in the long run, the player will lose 17 units per every 216 games he plays. ■

In the next example, we consider the simplest form of the theory of inheritance as developed by Gregor Mendel (1822–1884).

EXAMPLE 6d

Suppose that a particular trait (such as eye color or left-handedness) of a person is classified on the basis of one pair of genes, and suppose also that d represents a dominant gene and r a recessive gene. Thus, a person with dd genes is purely dominant, one with rr is purely recessive, and one with rd is hybrid. The purely dominant and the hybrid individuals are alike in appearance. Children receive 1 gene from each parent. If, with respect to a particular trait, 2 hybrid parents have a total of 4 children, what is the probability that 3 of the 4 children have the outward appearance of the dominant gene?

Solution. If we assume that each child is equally likely to inherit either of 2 genes from each parent, the probabilities that the child of 2 hybrid parents will have dd, rr, and rd pairs of genes are, respectively, $\frac{1}{4}, \frac{1}{4}$, and $\frac{1}{2}$. Hence, since an offspring will

have the outward appearance of the dominant gene if its gene pair is either *dd* or *rd*, it follows that the number of such children is binomially distributed with parameters $\left(4, \frac{3}{4}\right)$. Thus, the desired probability is

$$\binom{4}{3}\left(\frac{3}{4}\right)^3\left(\frac{1}{4}\right)^1 = \frac{27}{64}$$

■

EXAMPLE 6e

Consider a jury trial in which it takes 8 of the 12 jurors to convict the defendant; that is, in order for the defendant to be convicted, at least 8 of the jurors must vote him guilty. If we assume that jurors act independently and that, whether or not the defendant is guilty, each makes the right decision with probability θ, what is the probability that the jury renders a correct decision?

Solution. The problem, as stated, is incapable of solution, for there is not yet enough information. For instance, if the defendant is innocent, the probability of the jury's rendering a correct decision is

$$\sum_{i=5}^{12}\binom{12}{i}\theta^i(1-\theta)^{12-i}$$

whereas, if he is guilty, the probability of a correct decision is

$$\sum_{i=8}^{12}\binom{12}{i}\theta^i(1-\theta)^{12-i}$$

Therefore, if α represents the probability that the defendant is guilty, then, by conditioning on whether or not he is guilty, we obtain the probability that the jury renders a correct decision:

$$\alpha\sum_{i=8}^{12}\binom{12}{i}\theta^i(1-\theta)^{12-i} + (1-\alpha)\sum_{i=5}^{12}\binom{12}{i}\theta^i(1-\theta)^{12-i}$$

■

EXAMPLE 6f

A communication system consists of n components, each of which will, independently, function with probability p. The total system will be able to operate effectively if at least one-half of its components function.

(a) For what values of p is a 5-component system more likely to operate effectively than a 3-component system?
(b) In general, when is a $(2k+1)$-component system better than a $(2k-1)$-component system?

Solution. (a) Because the number of functioning components is a binomial random variable with parameters (n, p), it follows that the probability that a 5-component system will be effective is

$$\binom{5}{3}p^3(1-p)^2 + \binom{5}{4}p^4(1-p) + p^5$$

whereas the corresponding probability for a 3-component system is

$$\binom{3}{2} p^2(1 - p) + p^3$$

Hence, the 5-component system is better if

$$10p^3(1 - p)^2 + 5p^4(1 - p) + p^5 > 3p^2(1 - p) + p^3$$

which reduces to

$$3(p - 1)^2(2p - 1) > 0$$

or

$$p > \frac{1}{2}$$

(b) In general, a system with $2k + 1$ components will be better than one with $2k - 1$ components if (and only if) $p > \frac{1}{2}$. To prove this, consider a system of $2k + 1$ components and let X denote the number of the first $2k - 1$ that function. Then

$$\begin{aligned} P_{2k+1}(\text{effective}) = {} & P\{X \geq k + 1\} + P\{X = k\}(1 - (1 - p)^2) \\ & + P\{X = k - 1\}p^2 \end{aligned}$$

which follows because the $(2k + 1)$-component system will be effective if either

(i) $X \geq k + 1$;
(ii) $X = k$ and at least one of the remaining 2 components function; or
(iii) $X = k - 1$ and both of the next 2 components function.

Since

$$\begin{aligned} P_{2k-1}(\text{effective}) &= P\{X \geq k\} \\ &= P\{X = k\} + P\{X \geq k + 1\} \end{aligned}$$

we obtain

$$\begin{aligned} & P_{2k+1}(\text{effective}) - P_{2k-1}(\text{effective}) \\ & \quad = P\{X = k - 1\}p^2 - (1 - p)^2 P\{X = k\} \\ & \quad = \binom{2k - 1}{k - 1} p^{k-1}(1 - p)^k p^2 - (1 - p)^2 \binom{2k - 1}{k} p^k (1 - p)^{k-1} \\ & \quad = \binom{2k - 1}{k} p^k (1 - p)^k [p - (1 - p)] \text{ since } \binom{2k - 1}{k - 1} = \binom{2k - 1}{k} \\ & \quad > 0 \Leftrightarrow p > \frac{1}{2} \end{aligned}$$

■

4.6.1 Properties of Binomial Random Variables

We will now examine the properties of a binomial random variable with parameters n and p. To begin, let us compute its expected value and variance. Now,

$$\begin{aligned} E[X^k] &= \sum_{i=0}^{n} i^k \binom{n}{i} p^i (1-p)^{n-i} \\ &= \sum_{i=1}^{n} i^k \binom{n}{i} p^i (1-p)^{n-i} \end{aligned}$$

Using the identity

$$i\binom{n}{i} = n\binom{n-1}{i-1}$$

gives

$$\begin{aligned} E[X^k] &= np \sum_{i=1}^{n} i^{k-1} \binom{n-1}{i-1} p^{i-1} (1-p)^{n-i} \\ &= np \sum_{j=0}^{n-1} (j+1)^{k-1} \binom{n-1}{j} p^j (1-p)^{n-1-j} \quad \begin{matrix} \text{by letting} \\ j = i-1 \end{matrix} \\ &= npE[(Y+1)^{k-1}] \end{aligned}$$

where Y is a binomial random variable with parameters $n-1, p$. Setting $k=1$ in the preceding equation yields

$$E[X] = np$$

That is, the expected number of successes that occur in n independent trials when each is a success with probability p is equal to np. Setting $k=2$ in the preceding equation, and using the preceding formula for the expected value of a binomial random variable yields

$$\begin{aligned} E[X^2] &= npE[Y+1] \\ &= np[(n-1)p+1] \end{aligned}$$

Since $E[X] = np$, we obtain

$$\begin{aligned} \mathrm{Var}(X) &= E[X^2] - (E[X])^2 \\ &= np[(n-1)p+1] - (np)^2 \\ &= np(1-p) \end{aligned}$$

Summing up, we have shown the following:

If X is a binomial random variable with parameters n and p, then

$$\begin{aligned} E[X] &= np \\ \mathrm{Var}(X) &= np(1-p) \end{aligned}$$

The following proposition details how the binomial probability mass function first increases and then decreases.

Proposition 6.1. If X is a binomial random variable with parameters (n, p), where $0 < p < 1$, then as k goes from 0 to $n, P\{X = k\}$ first increases monotonically and then decreases monotonically, reaching its largest value when k is the largest integer less than or equal to $(n + 1)p$.

Proof. We prove the proposition by considering $P\{X = k\}/P\{X = k - 1\}$ and determining for what values of k it is greater or less than 1. Now,

$$\frac{P\{X = k\}}{P\{X = k - 1\}} = \frac{\dfrac{n!}{(n - k)!k!}p^k(1 - p)^{n-k}}{\dfrac{n!}{(n - k + 1)!(k - 1)!}p^{k-1}(1 - p)^{n-k+1}} = \frac{(n - k + 1)p}{k(1 - p)}$$

Hence, $P\{X = k\} \geq P\{X = k - 1\}$ if and only if

$$(n - k + 1)p \geq k(1 - p)$$

or, equivalently, if and only if

$$k \leq (n + 1)p$$

and the proposition is proved. ■

As an illustration of Proposition 6.1 consider Figure 4.5, the graph of the probability mass function of a binomial random variable with parameters $(10, \frac{1}{2})$.

EXAMPLE 6g

In a U.S. presidential election, the candidate who gains the maximum number of votes in a state is awarded the total number of electoral college votes allocated to

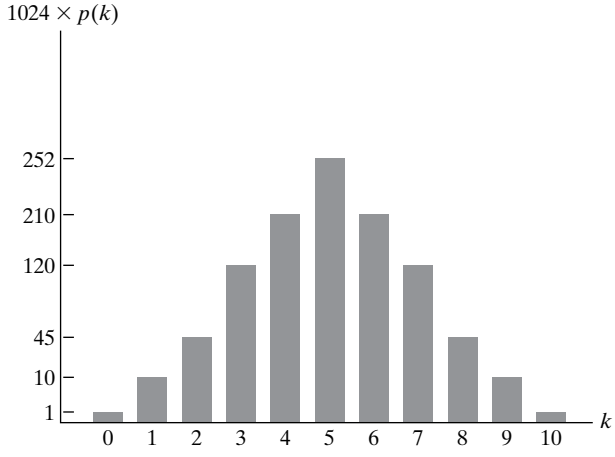


FIGURE 4.5 Graph of $p(k) = \binom{10}{k}\left(\frac{1}{2}\right)^{10}$

that state. The number of electoral college votes of a given state is roughly proportional to the population of that state—that is, a state with population n has roughly nc electoral votes. (Actually, it is closer to $nc + 2$, as a state is given an electoral vote for each member it has in the House of Representatives, with the number of such representatives being roughly proportional to the population of the state, and one electoral college vote for each of its two senators.) Let us determine the average power of a citizen in a state of size n in a close presidential election, where, by *average power in a close election*, we mean that a voter in a state of size $n = 2k + 1$ will be decisive if the other $n - 1$ voters split their votes evenly between the two candidates. (We are assuming here that n is odd, but the case where n is even is quite similar.) Because the election is close, we shall suppose that each of the other $n - 1 = 2k$ voters acts independently and is equally likely to vote for either candidate. Hence, the probability that a voter in a state of size $n = 2k + 1$ will make a difference to the outcome is the same as the probability that $2k$ tosses of a fair coin land heads and tails an equal number of times. That is,

$$\begin{aligned} &P\{\text{voter in state of size } 2k+1 \text{ makes a difference}\} \\ &\quad = \binom{2k}{k}\left(\frac{1}{2}\right)^k\left(\frac{1}{2}\right)^k \\ &\quad = \frac{(2k)!}{k!k!2^{2k}} \end{aligned}$$

To approximate the preceding equality, we make use of Stirling's approximation, which says that, for k large,

$$k! \sim k^{k+1/2}e^{-k}\sqrt{2\pi}$$

where we say that $a_k \sim b_k$ when the ratio a_k/b_k approaches 1 as k approaches ∞. Hence, it follows that

$$\begin{aligned} &P\{\text{voter in state of size } 2k+1 \text{ makes a difference}\} \\ &\sim \frac{(2k)^{2k+1/2}e^{-2k}\sqrt{2\pi}}{k^{2k+1}e^{-2k}(2\pi)2^{2k}} = \frac{1}{\sqrt{k\pi}} \end{aligned}$$

Because such a voter (if he or she makes a difference) will affect nc electoral votes, the expected number of electoral votes a voter in a state of size n will affect—or the voter's average power—is given by

$$\begin{aligned} \text{average power} &= ncP\{\text{makes a difference}\} \\ &\sim \frac{nc}{\sqrt{n\pi/2}} \\ &= c\sqrt{2n/\pi} \end{aligned}$$

Thus, the average power of a voter in a state of size n is proportional to the square root of n, showing that, in presidential elections, voters in large states have more power than do those in smaller states. ■

4.6.2 Computing the Binomial Distribution Function

Suppose that X is binomial with parameters (n, p). The key to computing its distribution function

$$P\{X \leq i\} = \sum_{k=0}^{i} \binom{n}{k} p^k (1 - p)^{n-k} \qquad i = 0, 1, \ldots, n$$

is to utilize the following relationship between $P\{X = k + 1\}$ and $P\{X = k\}$, which was established in the proof of Proposition 6.1:

$$P\{X = k + 1\} = \frac{p}{1 - p} \frac{n - k}{k + 1} P\{X = k\} \tag{6.3}$$

EXAMPLE 6h

Let X be a binomial random variable with parameters $n = 6, p = .4$. Then, starting with $P\{X = 0\} = (.6)^6$ and recursively employing Equation (6.3), we obtain

$$\begin{aligned}
P\{X = 0\} &= (.6)^6 \approx .0467 \\
P\{X = 1\} &= \frac{4}{6}\frac{6}{1} P\{X = 0\} \approx .1866 \\
P\{X = 2\} &= \frac{4}{6}\frac{5}{2} P\{X = 1\} \approx .3110 \\
P\{X = 3\} &= \frac{4}{6}\frac{4}{3} P\{X = 2\} \approx .2765 \\
P\{X = 4\} &= \frac{4}{6}\frac{3}{4} P\{X = 3\} \approx .1382 \\
P\{X = 5\} &= \frac{4}{6}\frac{2}{5} P\{X = 4\} \approx .0369 \\
P\{X = 6\} &= \frac{4}{6}\frac{1}{6} P\{X = 5\} \approx .0041
\end{aligned}$$

■

A computer program that utilizes the recursion (6.3) to compute the binomial distribution function is easily written. To compute $P\{X \leq i\}$, the program should first compute $P\{X = i\}$ and then use the recursion to successively compute $P\{X = i - 1\}, P\{X = i - 2\}$, and so on.

Historical Note

Independent trials having a common probability of success p were first studied by the Swiss mathematician Jacques Bernoulli (1654–1705). In his book *Ars Conjectandi (The Art of Conjecturing)*, published by his nephew Nicholas eight years after his death in 1713, Bernoulli showed that if the number of such trials were large, then the proportion of them that were successes would be close to p with a probability near 1.

Jacques Bernoulli was from the first generation of the most famous mathematical family of all time. Altogether, there were between 8 and 12 Bernoullis, spread over three generations, who made fundamental contributions to probability, statistics, and mathematics. One difficulty in knowing their exact number is the fact that several had the same name. (For example, two of the sons of Jacques's brother Jean

were named Jacques and Jean.) Another difficulty is that several of the Bernoullis were known by different names in different places. Our Jacques (sometimes written Jaques) was, for instance, also known as Jakob (sometimes written Jacob) and as James Bernoulli. But whatever their number, their influence and output were prodigious. Like the Bachs of music, the Bernoullis of mathematics were a family for the ages!

EXAMPLE 6i

If X is a binomial random variable with parameters $n = 100$ and $p = .75$, find $P\{X = 70\}$ and $P\{X \leq 70\}$.

Solution. The answer is shown here in Figure 4.6.

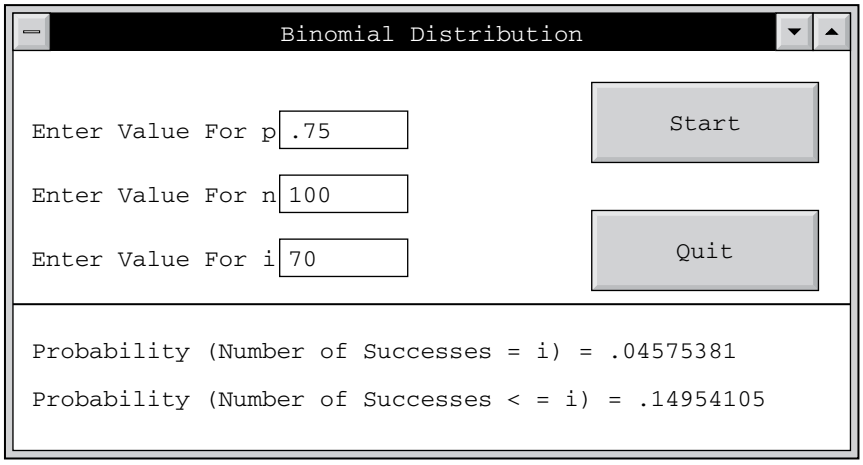


FIGURE 4.6

4.7 THE POISSON RANDOM VARIABLE

A random variable X that takes on one of the values $0, 1, 2, \ldots$ is said to be a *Poisson* random variable with parameter λ if, for some $\lambda > 0$,

$$p(i) = P\{X = i\} = e^{-\lambda}\frac{\lambda^i}{i!} \quad i = 0, 1, 2, \ldots \tag{7.1}$$

Equation (7.1) defines a probability mass function, since

$$\sum_{i=0}^{\infty} p(i) = e^{-\lambda}\sum_{i=0}^{\infty}\frac{\lambda^i}{i!} = e^{-\lambda}e^{\lambda} = 1$$

The Poisson probability distribution was introduced by Siméon Denis Poisson in a book he wrote regarding the application of probability theory to lawsuits, criminal trials, and the like. This book, published in 1837, was entitled *Recherches sur la probabilité des jugements en matière criminelle et en matière civile (Investigations into the Probability of Verdicts in Criminal and Civil Matters)*.

The Poisson random variable has a tremendous range of applications in diverse areas because it may be used as an approximation for a binomial random variable with parameters (n, p) when n is large and p is small enough so that np is of moderate

size. To see this, suppose that X is a binomial random variable with parameters (n, p), and let $\lambda = np$. Then

$$\begin{aligned} P\{X = i\} &= \frac{n!}{(n - i)!i!}p^i(1 - p)^{n-i} \\ &= \frac{n!}{(n - i)!i!}\left(\frac{\lambda}{n}\right)^i\left(1 - \frac{\lambda}{n}\right)^{n-i} \\ &= \frac{n(n - 1)\cdots(n - i + 1)}{n^i}\frac{\lambda^i}{i!}\frac{(1 - \lambda/n)^n}{(1 - \lambda/n)^i} \end{aligned}$$

Now, for n large and λ moderate,

$$\left(1 - \frac{\lambda}{n}\right)^n \approx e^{-\lambda} \quad \frac{n(n - 1)\cdots(n - i + 1)}{n^i} \approx 1 \quad \left(1 - \frac{\lambda}{n}\right)^i \approx 1$$

Hence, for n large and λ moderate,

$$P\{X = i\} \approx e^{-\lambda}\frac{\lambda^i}{i!}$$

In other words, if n independent trials, each of which results in a success with probability p, are performed, then, when n is large and p is small enough to make np moderate, the number of successes occurring is approximately a Poisson random variable with parameter $\lambda = np$. This value λ (which will later be shown to equal the expected number of successes) will usually be determined empirically.

Some examples of random variables that generally obey the Poisson probability law [that is, they obey Equation (7.1)] are as follows:

1. The number of misprints on a page (or a group of pages) of a book
2. The number of people in a community who survive to age 100
3. The number of wrong telephone numbers that are dialed in a day
4. The number of packages of dog biscuits sold in a particular store each day
5. The number of customers entering a post office on a given day
6. The number of vacancies occurring during a year in the federal judicial system
7. The number of α-particles discharged in a fixed period of time from some radioactive material

Each of the preceding, and numerous other random variables, are approximately Poisson for the same reason—namely, because of the Poisson approximation to the binomial. For instance, we can suppose that there is a small probability p that each letter typed on a page will be misprinted. Hence, the number of misprints on a page will be approximately Poisson with $\lambda = np$, where n is the number of letters on a page. Similarly, we can suppose that each person in a community has some small probability of reaching age 100. Also, each person entering a store may be thought of as having some small probability of buying a package of dog biscuits, and so on.

EXAMPLE 7a

Suppose that the number of typographical errors on a single page of this book has a Poisson distribution with parameter $\lambda = \frac{1}{2}$. Calculate the probability that there is at least one error on this page.

Solution. Letting X denote the number of errors on this page, we have

$$P\{X \geq 1\} = 1 - P\{X = 0\} = 1 - e^{-1/2} \approx .393$$

■

EXAMPLE 7b

Suppose that the probability that an item produced by a certain machine will be defective is .1. Find the probability that a sample of 10 items will contain at most 1 defective item.

Solution. The desired probability is $\binom{10}{0}(.1)^0(.9)^{10} + \binom{10}{1}(.1)^1(.9)^9 = .7361$, whereas the Poisson approximation yields the value $e^{-1} + e^{-1} \approx .7358$. ■

EXAMPLE 7c

Consider an experiment that consists of counting the number of α particles given off in a 1-second interval by 1 gram of radioactive material. If we know from past experience that, on the average, 3.2 such α particles are given off, what is a good approximation to the probability that no more than 2 α particles will appear?

Solution. If we think of the gram of radioactive material as consisting of a large number n of atoms, each of which has probability of $3.2/n$ of disintegrating and sending off an α particle during the second considered, then we see that, to a very close approximation, the number of α particles given off will be a Poisson random variable with parameter $\lambda = 3.2$. Hence, the desired probability is

$$\begin{aligned} P\{X \leq 2\} &= e^{-3.2} + 3.2e^{-3.2} + \frac{(3.2)^2}{2}e^{-3.2} \\ &\approx .3799 \end{aligned}$$

■

Before computing the expected value and variance of the Poisson random variable with parameter λ, recall that this random variable approximates a binomial random variable with parameters n and p when n is large, p is small, and $\lambda = np$. Since such a binomial random variable has expected value $np = \lambda$ and variance $np(1 - p) = \lambda(1 - p) \approx \lambda$ (since p is small), it would seem that both the expected value and the variance of a Poisson random variable would equal its parameter λ. We now verify this result:

$$\begin{aligned} E[X] &= \sum_{i=0}^{\infty} \frac{ie^{-\lambda}\lambda^i}{i!} \\ &= \lambda \sum_{i=1}^{\infty} \frac{e^{-\lambda}\lambda^{i-1}}{(i-1)!} \\ &= \lambda e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^j}{j!} \qquad \begin{matrix}\text{by letting} \\ j = i - 1\end{matrix} \\ &= \lambda \qquad \text{since } \sum_{j=0}^{\infty} \frac{\lambda^j}{j!} = e^{\lambda} \end{aligned}$$

Thus, the expected value of a Poisson random variable X is indeed equal to its parameter λ. To determine its variance, we first compute $E[X^2]$:

$$\begin{aligned} E[X^2] &= \sum_{i=0}^{\infty} \frac{i^2 e^{-\lambda}\lambda^i}{i!} \\ &= \lambda \sum_{i=1}^{\infty} \frac{i e^{-\lambda}\lambda^{i-1}}{(i-1)!} \\ &= \lambda \sum_{j=0}^{\infty} \frac{(j+1)e^{-\lambda}\lambda^j}{j!} \quad \begin{array}{l}\text{by letting} \\ j = i-1\end{array} \\ &= \lambda \left[\sum_{j=0}^{\infty} \frac{j e^{-\lambda}\lambda^j}{j!} + \sum_{j=0}^{\infty} \frac{e^{-\lambda}\lambda^j}{j!} \right] \\ &= \lambda(\lambda + 1) \end{aligned}$$

where the final equality follows because the first sum is the expected value of a Poisson random variable with parameter λ and the second is the sum of the probabilities of this random variable. Therefore, since we have shown that $E[X] = \lambda$, we obtain

$$\begin{aligned} \text{Var}(X) &= E[X^2] - (E[X])^2 \\ &= \lambda \end{aligned}$$

Hence, the expected value and variance of a Poisson random variable are both equal to its parameter λ.

We have shown that the Poisson distribution with parameter np is a very good approximation to the distribution of the number of successes in n independent trials when each trial has probability p of being a success, provided that n is large and p small. In fact, it remains a good approximation even when the trials are not independent, provided that their dependence is weak. For instance, recall the matching problem (Example 5m of Chapter 2) in which n men randomly select hats from a set consisting of one hat from each person. From the point of view of the number of men who select their own hat, we may regard the random selection as the result of n trials where we say that trial i is a success if person i selects his own hat, $i = 1, \ldots, n$. Defining the events $E_i, i = 1, \ldots, n$, by

$$E_i = \{\text{trial } i \text{ is a success}\}$$

it is easy to see that

$$P\{E_i\} = \frac{1}{n} \quad \text{and} \quad P\{E_i|E_j\} = \frac{1}{n-1}, \quad j \neq i$$

Thus, we see that, although the events $E_i, i = 1, \ldots, n$ are not independent, their dependence, for large n, appears to be weak. Because of this it seems reasonable to expect that the number of successes will approximately have a Poisson distribution with parameter $n \times 1/n = 1$, and indeed this is verified in Example 5m of Chapter 2.

For a second illustration of the strength of the Poisson approximation when the trials are weakly dependent, let us consider again the birthday problem presented in Example 5i of Chapter 2. In this example, we suppose that each of n people is equally likely to have any of the 365 days of the year as his or her birthday, and the problem

is to determine the probability that a set of n independent people all have different birthdays. A combinatorial argument was used to determine this probability, which was shown to be less than $\frac{1}{2}$ when $n = 23$.

We can approximate the preceding probability by using the Poisson approximation as follows: Imagine that we have a trial for each of the $\binom{n}{2}$ pairs of individuals i and $j, i \neq j$, and say that trial i, j is a success if persons i and j have the same birthday. If we let E_{ij} denote the event that trial i, j is a success, then, whereas the $\binom{n}{2}$ events $E_{ij}, 1 \leq i < j \leq n$, are not independent (see Theoretical Exercise 21), their dependence appears to be rather weak. (Indeed, these events are even *pairwise independent*, in that any 2 of the events E_{ij} and E_{kl} are independent—again, see Theoretical Exercise 21). Since $P(E_{ij}) = 1/365$, it is reasonable to suppose that the number of successes should approximately have a Poisson distribution with mean $\binom{n}{2} \Big/ 365 = n(n-1)/730$. Therefore,

$$\begin{aligned} P\{\text{no 2 people have the same birthday}\} &= P\{0 \text{ successes}\} \\ &\approx \exp\left\{\frac{-n(n-1)}{730}\right\} \end{aligned}$$

To determine the smallest integer n for which this probability is less than $\frac{1}{2}$, note that

$$\exp\left\{\frac{-n(n-1)}{730}\right\} \leq \frac{1}{2}$$

is equivalent to

$$\exp\left\{\frac{n(n-1)}{730}\right\} \geq 2$$

Taking logarithms of both sides, we obtain

$$\begin{aligned} n(n-1) &\geq 730 \log 2 \\ &\approx 505.997 \end{aligned}$$

which yields the solution $n = 23$, in agreement with the result of Example 5i of Chapter 2.

Suppose now that we wanted the probability that, among the n people, no 3 of them have their birthday on the same day. Whereas this now becomes a difficult combinatorial problem, it is a simple matter to obtain a good approximation. To begin, imagine that we have a trial for each of the $\binom{n}{3}$ triplets i, j, k, where $1 \leq i < j < k \leq n$, and call the i, j, k trial a success if persons i, j, and k all have their birthday on the same day. As before, we can then conclude that the number of successes is approximately a Poisson random variable with parameter

$$\begin{aligned} \binom{n}{3} P\{i, j, k \text{ have the same birthday}\} &= \binom{n}{3}\left(\frac{1}{365}\right)^2 \\ &= \frac{n(n-1)(n-2)}{6 \times (365)^2} \end{aligned}$$

Hence,

$$P\{\text{no 3 have the same birthday}\} \approx \exp\left\{\frac{-n(n-1)(n-2)}{799350}\right\}$$

This probability will be less than $\frac{1}{2}$ when n is such that

$$n(n-1)(n-2) \geq 799350 \log 2 \approx 554067.1$$

which is equivalent to $n \geq 84$. Thus, the approximate probability that at least 3 people in a group of size 84 or larger will have the same birthday exceeds $\frac{1}{2}$.

For the number of events to occur to approximately have a Poisson distribution, it is not essential that all the events have the same probability of occurrence, but only that all of these probabilities be small. The following is referred to as the *Poisson paradigm*.

Poisson Paradigm. Consider n events, with p_i equal to the probability that event i occurs, $i = 1, \ldots, n$. If all the p_i are "small" and the trials are either independent or at most "weakly dependent," then the number of these events that occur approximately has a Poisson distribution with mean $\sum_{i=1}^{n} p_i$.

Our next example not only makes use of the Poisson paradigm, but also illustrates a variety of the techniques we have studied so far.

EXAMPLE 7d Length of the longest run

A coin is flipped n times. Assuming that the flips are independent, with each one coming up heads with probability p, what is the probability that there is a string of k consecutive heads?

Solution. We will first use the Poisson paradigm to approximate this probability. Now, if, for $i = 1, \ldots, n-k+1$, we let H_i denote the event that flips $i, i+1, \ldots, i+k-1$ all land on heads, then the desired probability is that at least one of the events H_i occur. Because H_i is the event that, starting with flip i, the next k flips all land on heads, it follows that $P(H_i) = p^k$. Thus, when p^k is small, we might think that the number of the H_i that occur should have an approximate Poisson distribution. However, such is not the case, because, although the events all have small probabilities, some of their dependencies are too great for the Poisson distribution to be a good approximation. For instance, because the conditional probability that flips $2, \ldots, k+1$ are all heads given that flips $1, \ldots, k$ are all heads is equal to the probability that flip $k+1$ is a head, it follows that

$$P(H_2|H_1) = p$$

which is far greater than the unconditional probability of H_2.

The trick that enables us to use a Poisson approximation is to note that there will be a string of k consecutive heads either if there is such a string that is immediately followed by a tail or if the final k flips all land on heads. Consequently, for $i = 1, \ldots, n-k$, let E_i be the event that flips $i, \ldots, i+k-1$ are all heads and flip $i+k$ is a tail; also, let E_{n-k+1} be the event that flips $n-k+1, \ldots, n$ are all heads. Note that

$$P(E_i) = p^k(1-p), \quad i \leq n-k$$
$$P(E_{n-k+1}) = p^k$$

Thus, when p^k is small, each of the events E_i has a small probability of occurring. Moreover, for $i \neq j$, if the events E_i and E_j refer to nonoverlapping sequences of flips, then $P(E_i|E_j) = P(E_i)$; if they refer to overlapping sequences, then $P(E_i|E_j) = 0$. Hence, in both cases, the conditional probabilities are close to the unconditional ones, indicating that N, the number of the events E_i that occur, should have an approximate Poisson distribution with mean

$$E[N] = \sum_{i=1}^{n-k+1} P(E_i) = (n - k)p^k(1 - p) + p^k$$

Because there will not be a run of k heads if (and only if) $N = 0$, thus the preceding gives

$$P(\text{no head strings of length } k) = P(N = 0) \approx \exp\{-(n - k)p^k(1 - p) - p^k\}$$

If we let L_n denote the largest number of consecutive heads in the n flips, then, because L_n will be less than k if (and only if) there are no head strings of length k, the preceding equation can be written as

$$P\{L_n < k\} \approx \exp\{-(n - k)p^k(1 - p) - p^k\}$$

Now, let us suppose that the coin being flipped is fair; that is, suppose that $p = 1/2$. Then the preceding gives

$$P\{L_n < k\} \approx \exp\left\{-\frac{n - k + 2}{2^{k+1}}\right\} \approx \exp\left\{-\frac{n}{2^{k+1}}\right\}$$

where the final approximation supposes that $e^{\frac{k-2}{2^{k+1}}} \approx 1$ (that is, that $\frac{k-2}{2^{k+1}} \approx 0$). Let $j = \log_2 n$, and assume that j is an integer. For $k = j + i$,

$$\frac{n}{2^{k+1}} = \frac{n}{2^j 2^{i+1}} = \frac{1}{2^{i+1}}$$

Consequently,

$$P\{L_n < j + i\} \approx \exp\{-(1/2)^{i+1}\}$$

which implies that

$$\begin{aligned} P\{L_n = j + i\} &= P\{L_n < j + i + 1\} - P\{L_n < j + i\} \\ &\approx \exp\{-(1/2)^{i+2}\} - \exp\{-(1/2)^{i+1}\} \end{aligned}$$

For instance,

$$\begin{aligned} P\{L_n < j - 3\} &\approx e^{-4} \approx .0183 \\ P\{L_n = j - 3\} &\approx e^{-2} - e^{-4} \approx .1170 \\ P\{L_n = j - 2\} &\approx e^{-1} - e^{-2} \approx .2325 \\ P\{L_n = j - 1\} &\approx e^{-1/2} - e^{-1} \approx .2387 \\ P\{L_n = j\} &\approx e^{-1/4} - e^{-1/2} \approx .1723 \\ P\{L_n = j + 1\} &\approx e^{-1/8} - e^{-1/4} \approx .1037 \\ P\{L_n = j + 2\} &\approx e^{-1/16} - e^{-1/8} \approx .0569 \\ P\{L_n = j + 3\} &\approx e^{-1/32} - e^{-1/16} \approx .0298 \\ P\{L_n \geq j + 4\} &\approx 1 - e^{-1/32} \approx .0308 \end{aligned}$$

Thus, we observe the rather interesting fact that no matter how large n is, the length of the longest run of heads in a sequence of n flips of a fair coin will be within 2 of $\log_2(n) - 1$ with a probability approximately equal to .86.

We now derive an exact expression for the probability that there is a string of k consecutive heads when a coin that lands on heads with probability p is flipped n times. With the events $E_i, i = 1, \ldots, n - k + 1$, as defined earlier, and with L_n denoting, as before, the length of the longest run of heads,

$$P(L_n \geq k) = P(\text{there is a string of } k \text{ consecutive heads}) = P(\cup_{i=1}^{n-k+1} E_i)$$

The inclusion–exclusion identity for the probability of a union can be written as

$$P(\cup_{i=1}^{n-k+1} E_i) = \sum_{r=1}^{n-k+1} (-1)^{r+1} \sum_{i_1 < \cdots < i_r} P(E_{i_1} \cdots E_{i_r})$$

Let S_i denote the set of flip numbers to which the event E_i refers. (So, for instance, $S_1 = \{1, \ldots, k + 1\}$.) Now, consider one of the r-way intersection probabilities that does not include the event E_{n-k+1}. That is, consider $P(E_{i_1} \cdots E_{i_r})$ where $i_1 < \cdots < i_r < n - k + 1$. On the one hand, if there is any overlap in the sets $S_{i_1}, \ldots, S_{i_r}$ then this probability is 0. On the other hand, if there is no overlap, then the events $E_{i_1}, \ldots, E_{i_r}$ are independent. Therefore,

$$P(E_{i_1} \cdots E_{i_r}) = \begin{cases} 0, & \text{if there is any overlap in } S_{i_1}, \ldots, S_{i_r} \\ p^{rk}(1 - p)^r, & \text{if there is no overlap} \end{cases}$$

We must now determine the number of different choices of $i_1 < \cdots < i_r < n - k + 1$ for which there is no overlap in the sets $S_{i_1}, \ldots, S_{i_r}$. To do so, note first that each of the $S_{i_j}, j = 1, \ldots, r$, refer to $k + 1$ flips, so, without any overlap, they together refer to $r(k + 1)$ flips. Now consider any permutation of r identical letters a (one for each of the sets $S_{i_1}, \ldots, S_{i_{r-1}}$) and of $n - r(k + 1)$ identical letters b (one for each of the trials that are not part of any of $S_{i_1}, \ldots, S_{i_{r-1}}, S_{n-k+1}$). Interpret the number of b's before the first a as the number of flips before S_{i_1}, the number of b's between the first and second a as the number of flips between S_{i_1} and S_{i_2}, and so on, with the number of b's after the final a representing the number of flips after S_{i_r}. Because there are $\binom{n-rk}{r}$ permutations of r letters a and of $n - r(k + 1)$ letters b, with every such permutation corresponding (in a one-to-one fashion) to a different nonoverlapping choice, it follows that

$$\sum_{i_1 < \cdots < i_r < n-k+1} P(E_{i_1} \cdots E_{i_r}) = \binom{n - rk}{r} p^{rk}(1 - p)^r$$

We must now consider r-way intersection probabilities of the form

$$P(E_{i_1} \cdots E_{i_{r-1}} E_{n-k+1}),$$

where $i_1 < \cdots < i_{r-1} < n - k + 1$. Now, this probability will equal 0 if there is any overlap in $S_{i_1}, \ldots, S_{i_{r-1}}, S_{n-k}$; if there is no overlap, then the events of the intersection will be independent, so

$$P(E_{i_1} \cdots E_{i_{r-1}} E_{n-k+1}) = [p^k(1 - p)]^{r-1} p^k = p^{kr}(1 - p)^{r-1}$$

By a similar argument as before, the number of nonoverlapping sets $S_{i_1}, \ldots, S_{i_{r-1}}, S_{n-k}$ will equal the number of permutations of $r - 1$ letters a (one for each of the sets

$S_{i_1}, \ldots, S_{i_{r-1}}$) and of $n - (r-1)(k+1) - k = n - rk - (r-1)$ letters b (one for each of the trials that are not part of any of $S_{i_1}, \ldots, S_{i_{r-1}}, S_{n-k+1}$). Since there are $\binom{n-rk}{r-1}$ permutations of $r-1$ letters a and of $n - rk - (r-1)$ letters b, we have

$$\sum_{i_1 < \ldots < i_{r-1} < n-k+1} P(E_{i_1} \cdots E_{i_{r-1}} E_{n-k+1}) = \binom{n - rk}{r - 1} p^{kr} (1-p)^{r-1}$$

Putting it all together yields the exact expression, namely,

$$P(L_n \geq k) = \sum_{r=1}^{n-k+1} (-1)^{r+1} \left[\binom{n - rk}{r} + \frac{1}{p} \binom{n - rk}{r - 1} \right] p^{kr} (1-p)^r$$

where we utilize the convention that $\binom{m}{j} = 0$ if $m < j$.

From a computational point of view, a more efficient method for computing the desired probability than the use of the preceding identity is to derive a set of recursive equations. To do so, let A_n be the event that there is a string of k consecutive heads in a sequence of n flips of a fair coin, and let $P_n = P(A_n)$. We will derive a set of recursive equations for P_n by conditioning on when the first tail appears. For $j = 1, \ldots, k$, let F_j be the event that the first tail appears on flip j, and let H be the event that the first k flips are all heads. Because the events $F_1, \ldots, F_k, H$ are mutually exclusive and exhaustive (that is, exactly one of these events must occur), we have

$$P(A_n) = \sum_{j=1}^{k} P(A_n|F_j) P(F_j) + P(A_n|H) P(H)$$

Now, given that the first tail appears on flip j, where $j < k$, it follows that those j flips are wasted as far as obtaining a string of k heads in a row; thus, the conditional probability of this event is the probability that such a string will occur among the remaining $n - j$ flips. Therefore,

$$P(A_n|F_j) = P_{n-j}$$

Because $P(A_n|H) = 1$, the preceding equation gives

$$\begin{aligned} P_n &= P(A_n) \\ &= \sum_{j=1}^{k} P_{n-j} P(F_j) + P(H) \\ &= \sum_{j=1}^{k} P_{n-j} p^{j-1} (1-p) + p^k \end{aligned}$$

Starting with $P_j = 0, j < k$, and $P_k = p^k$, we can use the latter formula to recursively compute P_{k+1}, P_{k+2}, and so on, up to P_n. For instance, suppose we want the probability that there is a run of 2 consecutive heads when a fair coin is flipped 4 times. Then, with $k = 2$, we have $P_1 = 0, P_2 = (1/2)^2$. Because, when $p = 1/2$, the recursion becomes

$$P_n = \sum_{j=1}^{k} P_{n-j} (1/2)^j + (1/2)^k$$

we obtain

$$P_3 = P_2(1/2) + P_1(1/2)^2 + (1/2)^2 = 3/8$$

and

$$P_4 = P_3(1/2) + P_2(1/2)^2 + (1/2)^2 = 1/2$$

which is clearly true because there are 8 outcomes that result in a string of 2 consecutive heads: *hhhh*, *hhht*, *hhth*, *hthh*, *thhh*, *hhtt*, *thht*, and *tthh*. Each of these outcomes occurs with probability 1/16. ■

Another use of the Poisson probability distribution arises in situations where "events" occur at certain points in time. One example is to designate the occurrence of an earthquake as an event; another possibility would be for events to correspond to people entering a particular establishment (bank, post office, gas station, and so on); and a third possibility is for an event to occur whenever a war starts. Let us suppose that events are indeed occurring at certain (random) points of time, and let us assume that, for some positive constant λ, the following assumptions hold true:

1. The probability that exactly 1 event occurs in a given interval of length h is equal to $\lambda h + o(h)$, where $o(h)$ stands for any function $f(h)$ for which $\lim_{h\to 0} f(h)/h = 0$. [For instance, $f(h) = h^2$ is $o(h)$, whereas $f(h) = h$ is not.]
2. The probability that 2 or more events occur in an interval of length h is equal to $o(h)$.
3. For any integers $n, j_1, j_2, \ldots, j_n$ and any set of n nonoverlapping intervals, if we define E_i to be the event that exactly j_i of the events under consideration occur in the ith of these intervals, then events $E_1, E_2, \ldots, E_n$ are independent.

Loosely put, assumptions 1 and 2 state that, for small values of h, the probability that exactly 1 event occurs in an interval of size h equals λh plus something that is small compared with h, whereas the probability that 2 or more events occur is small compared with h. Assumption 3 states that whatever occurs in one interval has no (probability) effect on what will occur in other, nonoverlapping intervals.

We now show that, under assumptions 1, 2, and 3, the number of events occurring in any interval of length t is a Poisson random variable with parameter λt. To be precise, let us call the interval $[0, t]$ and denote the number of events occurring in that interval by $N(t)$. To obtain an expression for $P\{N(t) = k\}$, we start by breaking the interval $[0, t]$ into n nonoverlapping subintervals, each of length t/n (Figure 4.7).

Now,

$$\begin{aligned} P\{N(t) = k\} = {} & P\{k \text{ of the } n \text{ subintervals contain exactly 1 event} \\ & \quad \text{and the other } n - k \text{ contain 0 events}\} \\ & + P\{N(t) = k \text{ and at least 1 subinterval contains} \\ & \quad 2 \text{ or more events}\} \end{aligned} \tag{7.2}$$

The proceding equation holds because the event on the left side of Equation (7.2), that is, $\{N(t) = k\}$, is clearly equal to the union of the two mutually exclusive events

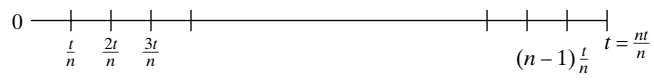


FIGURE 4.7

on the right side of the equation. Letting A and B denote the two mutually exclusive events on the right side of Equation (7.2), we have

$$\begin{aligned}
P(B) &\le P\{\text{at least one subinterval contains 2 or more events}\} \\
&= P\left(\bigcup_{i=1}^{n}\{i\text{th subinterval contains 2 or more events}\}\right) \\
&\le \sum_{i=1}^{n} P\{i\text{th subinterval contains 2 or more events}\} \quad \text{by Boole's inequality} \\
&= \sum_{i=1}^{n} o\left(\frac{t}{n}\right) \quad \text{by assumption 2} \\
&= no\left(\frac{t}{n}\right) \\
&= t\left[\frac{o(t/n)}{t/n}\right]
\end{aligned}$$

Now, in addition for any t, $t/n \to 0$ as $n \to \infty$, so $o(t/n)/(t/n) \to 0$ as $n \to \infty$, by the definition of $o(h)$. Hence,

$$P(B) \to 0 \quad \text{as} \quad n \to \infty \tag{7.3}$$

Moreover, since assumptions 1 and 2 imply that[†]

$$\begin{aligned}
&P\{0 \text{ events occur in an interval of length } h\} \\
&\quad = 1 - [\lambda h + o(h) + o(h)] = 1 - \lambda h - o(h)
\end{aligned}$$

we see from the independence assumption (number 3) that

$$\begin{aligned}
P(A) &= P\{k \text{ of the subintervals contain exactly 1 event and the other} \\
&\qquad n - k \text{ contain 0 events}\} \\
&= \binom{n}{k}\left[\frac{\lambda t}{n} + o\left(\frac{t}{n}\right)\right]^k \left[1 - \left(\frac{\lambda t}{n}\right) - o\left(\frac{t}{n}\right)\right]^{n-k}
\end{aligned}$$

However, since

$$n\left[\frac{\lambda t}{n} + o\left(\frac{t}{n}\right)\right] = \lambda t + t\left[\frac{o(t/n)}{t/n}\right] \to \lambda t \quad \text{as} \quad n \to \infty$$

it follows, by the same argument that verified the Poisson approximation to the binomial, that

$$P(A) \to e^{-\lambda t}\frac{(\lambda t)^k}{k!} \quad \text{as} \quad n \to \infty \tag{7.4}$$

Thus, from Equations (7.2), (7.3), and (7.4), by letting $n \to \infty$, we obtain

$$P\{N(t) = k\} = e^{-\lambda t}\frac{(\lambda t)^k}{k!} \quad k = 0, 1, \ldots \tag{7.5}$$

[†]The sum of two functions, both of which are $o(h)$, is also $o(h)$. This is so because if $\lim_{h\to 0} f(h)/h = \lim_{h\to 0} g(h)/h = 0$, then $\lim_{h\to 0}[f(h) + g(h)]/h = 0$.

Hence, if assumptions 1, 2, and 3 are satisfied, then the number of events occurring in any fixed interval of length t is a Poisson random variable with mean λt, and we say that the events occur in accordance with a Poisson process having rate λ. The value λ, which can be shown to equal the rate per unit time at which events occur, is a constant that must be empirically determined.

The preceding discussion explains why a Poisson random variable is usually a good approximation for such diverse phenomena as the following:

1. The number of earthquakes occurring during some fixed time span
2. The number of wars per year
3. The number of electrons emitted from a heated cathode during a fixed time period
4. The number of deaths, in a given period of time, of the policyholders of a life insurance company

EXAMPLE 7e

Suppose that earthquakes occur in the western portion of the United States in accordance with assumptions 1, 2, and 3, with $\lambda = 2$ and with 1 week as the unit of time. (That is, earthquakes occur in accordance with the three assumptions at a rate of 2 per week.)

(a) Find the probability that at least 3 earthquakes occur during the next 2 weeks.
(b) Find the probability distribution of the time, starting from now, until the next earthquake.

Solution. (a) From Equation (7.5), we have

$$\begin{aligned} P\{N(2) \geq 3\} &= 1 - P\{N(2) = 0\} - P\{N(2) = 1\} - P\{N(2) = 2\} \\ &= 1 - e^{-4} - 4e^{-4} - \frac{4^2}{2}e^{-4} \\ &= 1 - 13e^{-4} \end{aligned}$$

(b) Let X denote the amount of time (in weeks) until the next earthquake. Because X will be greater than t if and only if no events occur within the next t units of time, we have, from Equation (7.5),

$$P\{X > t\} = P\{N(t) = 0\} = e^{-\lambda t}$$

so the probability distribution function F of the random variable X is given by

$$\begin{aligned} F(t) = P\{X \leq t\} = 1 - P\{X > t\} &= 1 - e^{-\lambda t} \\ &= 1 - e^{-2t} \end{aligned}$$

■

4.7.1 Computing the Poisson Distribution Function

If X is Poisson with parameter λ, then

$$\frac{P\{X = i + 1\}}{P\{X = i\}} = \frac{e^{-\lambda}\lambda^{i+1}/(i+1)!}{e^{-\lambda}\lambda^{i}/i!} = \frac{\lambda}{i+1} \tag{7.6}$$

Starting with $P\{X = 0\} = e^{-\lambda}$, we can use (7.6) to compute successively

$$P\{X = 1\} = \lambda P\{X = 0\}$$
$$P\{X = 2\} = \frac{\lambda}{2} P\{X = 1\}$$
$$\vdots$$
$$P\{X = i + 1\} = \frac{\lambda}{i + 1} P\{X = i\}$$

The website includes a program that uses Equation (7.6) to compute Poisson probabilities.

EXAMPLE 7f

(a) Determine $P\{X \leq 90\}$ when X is Poisson with mean 100.
(b) Determine $P\{Y \leq 1075\}$ when Y is Poisson with mean 1000.

Solution. From the website, we obtain the solutions:

(a) $P\{X \leq 90\} \approx .1714$;
(b) $P\{Y \leq 1075\} \approx .9894$. ■

4.8 OTHER DISCRETE PROBABILITY DISTRIBUTIONS

4.8.1 The Geometric Random Variable

Suppose that independent trials, each having a probability $p, 0 < p < 1$, of being a success, are performed until a success occurs. If we let X equal the number of trials required, then

$$P\{X = n\} = (1 - p)^{n-1} p \qquad n = 1, 2, \ldots \tag{8.1}$$

Equation (8.1) follows because, in order for X to equal n, it is necessary and sufficient that the first $n - 1$ trials are failures and the nth trial is a success. Equation (8.1) then follows, since the outcomes of the successive trials are assumed to be independent.

Since

$$\sum_{n=1}^{\infty} P\{X = n\} = p \sum_{n=1}^{\infty} (1 - p)^{n-1} = \frac{p}{1 - (1 - p)} = 1$$

it follows that, with probability 1, a success will eventually occur. Any random variable X whose probability mass function is given by Equation (8.1) is said to be a *geometric* random variable with parameter p.

EXAMPLE 8a

An urn contains N white and M black balls. Balls are randomly selected, one at a time, until a black one is obtained. If we assume that each ball selected is replaced before the next one is drawn, what is the probability that

(a) exactly n draws are needed?
(b) at least k draws are needed?

Solution. If we let X denote the number of draws needed to select a black ball, then X satisfies Equation (8.1) with $p = M/(M + N)$. Hence,

(a)

$$P\{X = n\} = \left(\frac{N}{M + N}\right)^{n-1} \frac{M}{M + N} = \frac{MN^{n-1}}{(M + N)^n}$$

(b)

$$\begin{aligned} P\{X \geq k\} &= \frac{M}{M + N} \sum_{n=k}^{\infty} \left(\frac{N}{M + N}\right)^{n-1} \\ &= \left(\frac{M}{M + N}\right) \left(\frac{N}{M + N}\right)^{k-1} \Big/ \left[1 - \frac{N}{M + N}\right] \\ &= \left(\frac{N}{M + N}\right)^{k-1} \end{aligned}$$

Of course, part (b) could have been obtained directly, since the probability that at least k trials are necessary to obtain a success is equal to the probability that the first $k - 1$ trials are all failures. That is, for a geometric random variable,

$$P\{X \geq k\} = (1 - p)^{k-1}$$

■

EXAMPLE 8b

Find the expected value of a geometric random variable.

Solution. With $q = 1 - p$, we have

$$\begin{aligned} E[X] &= \sum_{i=1}^{\infty} iq^{i-1}p \\ &= \sum_{i=1}^{\infty} (i - 1 + 1)q^{i-1}p \\ &= \sum_{i=1}^{\infty} (i - 1)q^{i-1}p + \sum_{i=1}^{\infty} q^{i-1}p \\ &= \sum_{j=0}^{\infty} jq^j p + 1 \\ &= q \sum_{j=1}^{\infty} jq^{j-1}p + 1 \\ &= qE[X] + 1 \end{aligned}$$

Hence,

$$pE[X] = 1$$

yielding the result

$$E[X] = \frac{1}{p}$$

In other words, if independent trials having a common probability p of being successful are performed until the first success occurs, then the expected number of required trials equals $1/p$. For instance, the expected number of rolls of a fair die that it takes to obtain the value 1 is 6. ■

EXAMPLE 8c

Find the variance of a geometric random variable.

Solution. To determine $\text{Var}(X)$, let us first compute $E[X^2]$. With $q = 1 - p$, we have

$$\begin{aligned}
E[X^2] &= \sum_{i=1}^{\infty} i^2 q^{i-1} p \\
&= \sum_{i=1}^{\infty} (i - 1 + 1)^2 q^{i-1} p \\
&= \sum_{i=1}^{\infty} (i - 1)^2 q^{i-1} p + \sum_{i=1}^{\infty} 2(i - 1) q^{i-1} p + \sum_{i=1}^{\infty} q^{i-1} p \\
&= \sum_{j=0}^{\infty} j^2 q^j p + 2 \sum_{j=1}^{\infty} j q^j p + 1 \\
&= qE[X^2] + 2qE[X] + 1
\end{aligned}$$

Using $E[X] = 1/p$, the equation for $E[X^2]$ yields

$$pE[X^2] = \frac{2q}{p} + 1$$

Hence,

$$E[X^2] = \frac{2q + p}{p^2} = \frac{q + 1}{p^2}$$

giving the result

$$\text{Var}(X) = \frac{q + 1}{p^2} - \frac{1}{p^2} = \frac{q}{p^2} = \frac{1 - p}{p^2}$$

■

4.8.2 The Negative Binomial Random Variable

Suppose that independent trials, each having probability $p, 0 < p < 1$, of being a success are performed until a total of r successes is accumulated. If we let X equal the number of trials required, then

$$P\{X = n\} = \binom{n - 1}{r - 1} p^r (1 - p)^{n-r} \quad n = r, r + 1, \ldots \tag{8.2}$$

Equation (8.2) follows because, in order for the rth success to occur at the nth trial, there must be $r - 1$ successes in the first $n - 1$ trials and the nth trial must be a success. The probability of the first event is

$$\binom{n - 1}{r - 1} p^{r-1} (1 - p)^{n-r}$$

and the probability of the second is p; thus, by independence, Equation (8.2) is established. To verify that a total of r successes must eventually be accumulated, either we can prove analytically that

$$\sum_{n=r}^{\infty} P\{X = n\} = \sum_{n=r}^{\infty} \binom{n-1}{r-1} p^r (1-p)^{n-r} = 1 \tag{8.3}$$

or we can give a probabilistic argument as follows: The number of trials required to obtain r successes can be expressed as $Y_1 + Y_2 + \cdots + Y_r$, where Y_1 equals the number of trials required for the first success, Y_2 the number of additional trials after the first success until the second success occurs, Y_3 the number of additional trials until the third success, and so on. Because the trials are independent and all have the same probability of success, it follows that $Y_1, Y_2, \ldots, Y_r$ are all geometric random variables. Hence, each is finite with probability 1, so $\sum_{i=1}^{r} Y_i$ must also be finite, establishing Equation (8.3).

Any random variable X whose probability mass function is given by Equation (8.2) is said to be a *negative binomial* random variable with parameters (r, p). Note that a geometric random variable is just a negative binomial with parameter $(1, p)$.

In the next example, we use the negative binomial to obtain another solution of the problem of the points.

EXAMPLE 8d

If independent trials, each resulting in a success with probability p, are performed, what is the probability of r successes occurring before m failures?

Solution. The solution will be arrived at by noting that r successes will occur before m failures if and only if the rth success occurs no later than the $(r + m - 1)$th trial. This follows because if the rth success occurs before or at the $(r + m - 1)$th trial, then it must have occurred before the mth failure, and conversely. Hence, from Equation (8.2), the desired probability is

$$\sum_{n=r}^{r+m-1} \binom{n-1}{r-1} p^r (1-p)^{n-r}$$

■

EXAMPLE 8e The Banach match problem

At all times, a pipe-smoking mathematician carries 2 matchboxes—1 in his left-hand pocket and 1 in his right-hand pocket. Each time he needs a match, he is equally likely to take it from either pocket. Consider the moment when the mathematician first discovers that one of his matchboxes is empty. If it is assumed that both matchboxes initially contained N matches, what is the probability that there are exactly k matches, $k = 0, 1, \ldots, N$, in the other box?

Solution. Let E denote the event that the mathematician first discovers that the right-hand matchbox is empty and that there are k matches in the left-hand box at the time. Now, this event will occur if and only if the $(N + 1)$th choice of the right-hand matchbox is made at the $(N + 1 + N - k)$th trial. Hence, from Equation (8.2) (with $p = \frac{1}{2}, r = N + 1,$ and $n = 2N - k + 1$), we see that

$$P(E) = \binom{2N-k}{N} \left(\frac{1}{2}\right)^{2N-k+1}$$

Since there is an equal probability that it is the left-hand box that is first discovered to be empty and there are k matches in the right-hand box at that time, the desired result is

$$2P(E) = \binom{2N - k}{N} \left(\frac{1}{2}\right)^{2N-k}$$

■

EXAMPLE 8f

Compute the expected value and the variance of a negative binomial random variable with parameters r and p.

Solution. We have

$$\begin{aligned}
E[X^k] &= \sum_{n=r}^{\infty} n^k \binom{n-1}{r-1} p^r (1-p)^{n-r} \\
&= \frac{r}{p} \sum_{n=r}^{\infty} n^{k-1} \binom{n}{r} p^{r+1} (1-p)^{n-r} \quad \text{since} \quad n\binom{n-1}{r-1} = r\binom{n}{r} \\
&= \frac{r}{p} \sum_{m=r+1}^{\infty} (m-1)^{k-1} \binom{m-1}{r} p^{r+1} (1-p)^{m-(r+1)} \quad \text{by setting } m = n+1 \\
&= \frac{r}{p} E[(Y-1)^{k-1}]
\end{aligned}$$

where Y is a negative binomial random variable with parameters $r + 1, p$. Setting $k = 1$ in the preceding equation yields

$$E[X] = \frac{r}{p}$$

Setting $k = 2$ in the equation for $E[X^k]$ and using the formula for the expected value of a negative binomial random variable gives

$$\begin{aligned}
E[X^2] &= \frac{r}{p} E[Y - 1] \\
&= \frac{r}{p} \left(\frac{r+1}{p} - 1\right)
\end{aligned}$$

Therefore,

$$\begin{aligned}
\text{Var}(X) &= \frac{r}{p} \left(\frac{r+1}{p} - 1\right) - \left(\frac{r}{p}\right)^2 \\
&= \frac{r(1-p)}{p^2}
\end{aligned}$$

■

Thus, from Example 8f, if independent trials, each of which is a success with probability p, are performed, then the expected value and variance of the number of trials that it takes to amass r successes is r/p and $r(1-p)/p^2$, respectively.

Since a geometric random variable is just a negative binomial with parameter $r = 1$, it follows from the preceding example that the variance of a geometric random variable with parameter p is equal to $(1-p)/p^2$, which checks with the result of Example 8c.

EXAMPLE 8g

Find the expected value and the variance of the number of times one must throw a die until the outcome 1 has occurred 4 times.

Solution. Since the random variable of interest is a negative binomial with parameters $r = 4$ and $p = \frac{1}{6}$, it follows that

$$E[X] = 24$$

$$\text{Var}(X) = \frac{4\left(\frac{5}{6}\right)}{\left(\frac{1}{6}\right)^2} = 120$$

■

4.8.3 The Hypergeometric Random Variable

Suppose that a sample of size n is to be chosen randomly (without replacement) from an urn containing N balls, of which m are white and $N - m$ are black. If we let X denote the number of white balls selected, then

$$P\{X = i\} = \frac{\binom{m}{i}\binom{N-m}{n-i}}{\binom{N}{n}} \quad i = 0, 1, \ldots, n \tag{8.4}$$

A random variable X whose probability mass function is given by Equation (8.4) for some values of n, N, m is said to be a *hypergeometric* random variable.

Remark. Although we have written the hypergeometric probability mass function with i going from 0 to n, $P\{X = i\}$ will actually be 0, unless i satisfies the inequalities $n - (N - m) \leq i \leq \min(n, m)$. However, Equation (8.4) is always valid because of our convention that $\binom{r}{k}$ is equal to 0 when either $k < 0$ or $r < k$. ■

EXAMPLE 8h

An unknown number, say, N, of animals inhabit a certain region. To obtain some information about the size of the population, ecologists often perform the following experiment: They first catch a number, say, m, of these animals, mark them in some manner, and release them. After allowing the marked animals time to disperse throughout the region, a new catch of size, say, n, is made. Let X denote the number of marked animals in this second capture. If we assume that the population of animals in the region remained fixed between the time of the two catches and that each time an animal was caught it was equally likely to be any of the remaining uncaught animals, it follows that X is a hypergeometric random variable such that

$$P\{X = i\} = \frac{\binom{m}{i}\binom{N-m}{n-i}}{\binom{N}{n}} \equiv P_i(N)$$

Suppose now that X is observed to equal i. Then, since $P_i(N)$ represents the probability of the observed event when there are actually N animals present in the region, it would appear that a reasonable estimate of N would be the value of N that maximizes $P_i(N)$. Such an estimate is called a *maximum likelihood* estimate. (See Theoretical Exercises 13 and 18 for other examples of this type of estimation procedure.)

The maximization of $P_i(N)$ can be done most simply by first noting that

$$\frac{P_i(N)}{P_i(N-1)} = \frac{(N-m)(N-n)}{N(N-m-n+i)}$$

Now, the preceding ratio is greater than 1 if and only if

$$(N-m)(N-n) \geq N(N-m-n+i)$$

or, equivalently, if and only if

$$N \leq \frac{mn}{i}$$

Thus, $P_i(N)$ is first increasing and then decreasing, and reaches its maximum value at the largest integral value not exceeding mn/i. This value is the maximum likelihood estimate of N. For example, suppose that the initial catch consisted of $m = 50$ animals, which are marked and then released. If a subsequent catch consists of $n = 40$ animals of which $i = 4$ are marked, then we would estimate that there are some 500 animals in the region. (Note that the preceding estimate could also have been obtained by assuming that the proportion of marked animals in the region, m/N, is approximately equal to the proportion of marked animals in our second catch, i/n.) ■

EXAMPLE 8i

A purchaser of electrical components buys them in lots of size 10. It is his policy to inspect 3 components randomly from a lot and to accept the lot only if all 3 are nondefective. If 30 percent of the lots have 4 defective components and 70 percent have only 1, what proportion of lots does the purchaser reject?

Solution. Let A denote the event that the purchaser accepts a lot. Now,

$$\begin{aligned} P(A) &= P(A|\text{lot has 4 defectives})\frac{3}{10} + P(A|\text{lot has 1 defective})\frac{7}{10} \\ &= \frac{\binom{4}{0}\binom{6}{3}}{\binom{10}{3}}\left(\frac{3}{10}\right) + \frac{\binom{1}{0}\binom{9}{3}}{\binom{10}{3}}\left(\frac{7}{10}\right) \\ &= \frac{54}{100} \end{aligned}$$

Hence, 46 percent of the lots are rejected. ■

If n balls are randomly chosen without replacement from a set of N balls of which the fraction $p = m/N$ is white, then the number of white balls selected is hypergeometric. Now, it would seem that when m and N are large in relation to n, it shouldn't make much difference whether the selection is being done with or without replacement, because, no matter which balls have previously been selected, when m and N are large, each additional selection will be white with a probability approximately equal to p. In other words, it seems intuitive that when m and N are large in relation to n, the probability mass function of X should approximately be that of a binomial random variable with parameters n and p. To verify this intuition, note that if X is hypergeometric, then, for $i \leq n$,

$$
\begin{aligned}
P\{X = i\} &= \frac{\binom{m}{i}\binom{N-m}{n-i}}{\binom{N}{n}} \\
&= \frac{m!}{(m-i)!\,i!}\,\frac{(N-m)!}{(N-m-n+i)!\,(n-i)!}\,\frac{(N-n)!\,n!}{N!} \\
&= \binom{n}{i}\frac{m}{N}\frac{m-1}{N-1}\cdots\frac{m-i+1}{N-i+1}\frac{N-m}{N-i}\frac{N-m-1}{N-i-1} \\
&\quad \cdots\frac{N-m-(n-i-1)}{N-i-(n-i-1)} \\
&\approx \binom{n}{i}p^i(1-p)^{n-i} \qquad \begin{array}{l}\text{when } p = m/N \text{ and } m \text{ and } N \text{ are} \\ \text{large in relation to } n \text{ and } i\end{array}
\end{aligned}
$$

EXAMPLE 8j

Determine the expected value and the variance of X, a hypergeometric random variable with parameters n, N, and m.

Solution.

$$
\begin{aligned}
E[X^k] &= \sum_{i=0}^{n} i^k P\{X = i\} \\
&= \sum_{i=1}^{n} i^k \binom{m}{i}\binom{N-m}{n-i} \Big/ \binom{N}{n}
\end{aligned}
$$

Using the identities

$$
i\binom{m}{i} = m\binom{m-1}{i-1} \quad \text{and} \quad n\binom{N}{n} = N\binom{N-1}{n-1}
$$

we obtain

$$
\begin{aligned}
E[X^k] &= \frac{nm}{N}\sum_{i=1}^{n} i^{k-1}\binom{m-1}{i-1}\binom{N-m}{n-i} \Big/ \binom{N-1}{n-1} \\
&= \frac{nm}{N}\sum_{j=0}^{n-1}(j+1)^{k-1}\binom{m-1}{j}\binom{N-m}{n-1-j} \Big/ \binom{N-1}{n-1} \\
&= \frac{nm}{N}E[(Y+1)^{k-1}]
\end{aligned}
$$

where Y is a hypergeometric random variable with parameters $n-1$, $N-1$, and $m-1$. Hence, upon setting $k = 1$, we have

$$
E[X] = \frac{nm}{N}
$$

In words, if n balls are randomly selected from a set of N balls, of which m are white, then the expected number of white balls selected is nm/N.

Upon setting $k = 2$ in the equation for $E[X^k]$, we obtain

$$\begin{aligned} E[X^2] &= \frac{nm}{N} E[Y + 1] \\ &= \frac{nm}{N}\left[\frac{(n-1)(m-1)}{N-1} + 1\right] \end{aligned}$$

where the final equality uses our preceding result to compute the expected value of the hypergeometric random variable Y.

Because $E[X] = nm/N$, we can conclude that

$$\mathrm{Var}(X) = \frac{nm}{N}\left[\frac{(n-1)(m-1)}{N-1} + 1 - \frac{nm}{N}\right]$$

Letting $p = m/N$ and using the identity

$$\frac{m-1}{N-1} = \frac{Np-1}{N-1} = p - \frac{1-p}{N-1}$$

shows that

$$\begin{aligned} \mathrm{Var}(X) &= np[(n-1)p - (n-1)\frac{1-p}{N-1} + 1 - np] \\ &= np(1-p)(1 - \frac{n-1}{N-1}) \end{aligned}$$

■

Remark. We have shown in Example 8j that if n balls are randomly selected without replacement from a set of N balls, of which the fraction p are white, then the expected number of white balls chosen is np. In addition, if N is large in relation to n [so that $(N-n)/(N-1)$ is approximately equal to 1], then

$$\mathrm{Var}(X) \approx np(1-p)$$

In other words, $E[X]$ is the same as when the selection of the balls is done with replacement (so that the number of white balls is binomial with parameters n and p), and if the total collection of balls is large, then $\mathrm{Var}(X)$ is approximately equal to what it would be if the selection were done with replacement. This is, of course, exactly what we would have guessed, given our earlier result that when the number of balls in the urn is large, the number of white balls chosen approximately has the mass function of a binomial random variable. ■

4.8.4 The Zeta (or Zipf) Distribution

A random variable is said to have a zeta (sometimes called the Zipf) distribution if its probability mass function is given by

$$P\{X = k\} = \frac{C}{k^{\alpha+1}} \quad k = 1, 2, \ldots$$

for some value of $\alpha > 0$. Since the sum of the foregoing probabilities must equal 1, it follows that

$$C = \left[\sum_{k=1}^{\infty} \left(\frac{1}{k}\right)^{\alpha+1}\right]^{-1}$$

The zeta distribution owes its name to the fact that the function

$$\zeta(s) = 1 + \left(\frac{1}{2}\right)^s + \left(\frac{1}{3}\right)^s + \cdots + \left(\frac{1}{k}\right)^s + \cdots$$

is known in mathematical disciplines as the Riemann zeta function (after the German mathematician G. F. B. Riemann).

The zeta distribution was used by the Italian economist V. Pareto to describe the distribution of family incomes in a given country. However, it was G. K. Zipf who applied zeta distribution to a wide variety of problems in different areas and, in doing so, popularized its use.

4.9 EXPECTED VALUE OF SUMS OF RANDOM VARIABLES

A very important property of expectations is that the expected value of a sum of random variables is equal to the sum of their expectations. In this section, we will prove this result under the assumption that the set of possible values of the probability experiment—that is, the sample space S—is either finite or countably infinite. Although the result is true without this assumption (and a proof is outlined in the theoretical exercises), not only will the assumption simplify the argument, but it will also result in an enlightening proof that will add to our intuition about expectations. So, for the remainder of this section, suppose that the sample space S is either a finite or a countably infinite set.

For a random variable X, let $X(s)$ denote the value of X when $s \in S$ is the outcome of the experiment. Now, if X and Y are both random variables, then so is their sum. That is, $Z = X + Y$ is also a random variable. Moreover, $Z(s) = X(s) + Y(s)$.

EXAMPLE 9a

Suppose that the experiment consists of flipping a coin 5 times, with the outcome being the resulting sequence of heads and tails. Suppose X is the number of heads in the first 3 flips and Y is the number of heads in the final 2 flips. Let $Z = X + Y$. Then, for instance, for the outcome $s = (h, t, h, t, h)$,

$$\begin{aligned} X(s) &= 2 \\ Y(s) &= 1 \\ Z(s) &= X(s) + Y(s) = 3 \end{aligned}$$

meaning that the outcome (h, t, h, t, h) results in 2 heads in the first three flips, 1 head in the final two flips, and a total of 3 heads in the five flips. ■

Let $p(s) = P(\{s\})$ be the probability that s is the outcome of the experiment. Because we can write any event A as the finite or countably infinite union of the mutually exclusive events $\{s\}, s \in A$, it follows by the axioms of probability that

$$P(A) = \sum_{s \in A} p(s)$$

When $A = S$, the preceding equation gives

$$1 = \sum_{s \in S} p(s)$$

Now, let X be a random variable, and consider $E[X]$. Because $X(s)$ is the value of X when s is the outcome of the experiment, it seems intuitive that $E[X]$—the weighted average of the possible values of X, with each value weighted by the probability that X assumes that value—should equal a weighted average of the values $X(s), s \in S$, with $X(s)$ weighted by the probability that s is the outcome of the experiment. We now prove this intuition.

Proposition 9.1.

$$E[X] = \sum_{s \in S} X(s)\, p(s)$$

Proof. Suppose that the distinct values of X are $x_i, i \geq 1$. For each i, let S_i be the event that X is equal to x_i. That is, $S_i = \{s : X(s) = x_i\}$. Then,

$$\begin{aligned} E[X] &= \sum_i x_i P\{X = x_i\} \\ &= \sum_i x_i P(S_i) \\ &= \sum_i x_i \sum_{s \in S_i} p(s) \\ &= \sum_i \sum_{s \in S_i} x_i p(s) \\ &= \sum_i \sum_{s \in S_i} X(s) p(s) \\ &= \sum_{s \in S} X(s) p(s) \end{aligned}$$

where the final equality follows because $S_1, S_2, \ldots$ are mutually exclusive events whose union is S. □

EXAMPLE 9b

Suppose that two independent flips of a coin that comes up heads with probability p are made, and let X denote the number of heads obtained. Because

$$\begin{aligned} P(X = 0) &= P(t,t) = (1 - p)^2, \\ P(X = 1) &= P(h,t) + P(t,h) = 2p(1 - p) \\ P(X = 2) &= P(h,h) = p^2 \end{aligned}$$

it follows from the definition of expected value that

$$E[X] = 0 \cdot (1 - p)^2 + 1 \cdot 2p(1 - p) + 2 \cdot p^2 = 2p$$

which agrees with

$$\begin{aligned} E[X] &= X(h,h)p^2 + X(h,t)p(1 - p) + X(t,h)(1 - p)p + X(t,t)(1 - p)^2 \\ &= 2p^2 + p(1 - p) + (1 - p)p \\ &= 2p \end{aligned}$$

■

We now prove the important and useful result that the expected value of a sum of random variables is equal to the sum of their expectations.

Corollary 9.2. For random variables $X_1, X_2, \ldots, X_n$,

$$E\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} E[X_i]$$

Proof. Let $Z = \sum_{i=1}^{n} X_i$. Then, by Proposition 9.1,

$$\begin{aligned} E[Z] &= \sum_{s \in S} Z(s)p(s) \\ &= \sum_{s \in S} \big(X_1(s) + X_2(s) + \ldots + X_n(s)\big)\, p(s) \\ &= \sum_{s \in S} X_1(s)p(s) + \sum_{s \in S} X_2(s)p(s) + \ldots + \sum_{s \in S} X_n(s)p(s) \\ &= E[X_1] + E[X_2] + \ldots + E[X_n] \end{aligned}$$

■

EXAMPLE 9c

Find the expected value of the sum obtained when n fair dice are rolled.

Solution. Let X be the sum. We will compute $E[X]$ by using the representation

$$X = \sum_{i=1}^{n} X_i$$

where X_i is the upturned value on die i. Because X_i is equally likely to be any of the values from 1 to 6, it follows that

$$E[X_i] = \sum_{i=1}^{6} i(1/6) = 21/6 = 7/2$$

which yields the result

$$E[X] = E\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} E[X_i] = 3.5\, n$$

■

EXAMPLE 9d

Find the expected total number of successes that result from n trials when trial i is a success with probability p_i, $i = 1, \ldots, n$.

Solution. Letting

$$X_i = \begin{cases} 1, & \text{if trial } i \text{ is a success} \\ 0, & \text{if trial } i \text{ is a failure} \end{cases}$$

we have the representation

$$X = \sum_{i=1}^{n} X_i$$

Consequently,

$$E[X] = \sum_{i=1}^{n} E[X_i] = \sum_{i=1}^{n} p_i$$

Note that this result does not require that the trials be independent. It includes as a special case the expected value of a binomial random variable, which assumes independent trials and all $p_i = p$, and thus has mean np. It also gives the expected value of a hypergeometric random variable representing the number of white balls selected when n balls are randomly selected, without replacement, from an urn of N balls of which m are white. We can interpret the hypergeometric as representing the number of successes in n trials, where trial i is said to be a success if the ith ball selected is white. Because the ith ball selected is equally likely to be any of the N balls and thus has probability m/N of being white, it follows that the hypergeometric is the number of successes in n trials in which each trial is a success with probability $p = m/N$. Hence, even though these hypergeometric trials are dependent, it follows from the result of Example 9d that the expected value of the hypergeometric is $np = nm/N$. ■

EXAMPLE 9e

Derive an expression for the variance of the number of successful trials in Example 9*d*, and apply it to obtain the variance of a binomial random variable with parameters n and p, and of a hypergeometric random variable equal to the number of white balls chosen when n balls are randomly chosen from an urn containing N balls of which m are white.

Solution. Letting X be the number of successful trials, and using the same representation for X—namely, $X = \sum_{i=1}^{n} X_i$—as in the previous example, we have

$$\begin{aligned}
E[X^2] &= E\left[\left(\sum_{i=1}^{n} X_i\right)\left(\sum_{j=1}^{n} X_j\right)\right] \\
&= E\left[\sum_{i=1}^{n} X_i\left(X_i + \sum_{j\neq i} X_j\right)\right] \\
&= E\left[\sum_{i=1}^{n} X_i^2 + \sum_{i=1}^{n}\sum_{j\neq i} X_i X_j\right] \\
&= \sum_{i=1}^{n} E[X_i^2] + \sum_{i=1}^{n}\sum_{j\neq i} E[X_i X_j] \\
&= \sum_{i} p_i + \sum_{i=1}^{n}\sum_{j\neq i} E[X_i X_j] \qquad (9.1)
\end{aligned}$$

where the final equation used that $X_i^2 = X_i$. However, because the possible values of both X_i and X_j are 0 or 1, it follows that

$$X_i X_j = \begin{cases} 1, & \text{if } X_i = 1, X_j = 1 \\ 0, & \text{otherwise} \end{cases}$$

Hence,

$$E[X_iX_j] = P\{X_i = 1, X_j = 1\} = P(\text{trials } i \text{ and } j \text{ are successes})$$

Now, on the one hand, if X is binomial, then, for $i \neq j$, the results of trial i and trial j are independent, with each being a success with probability p. Therefore,

$$E[X_iX_j] = p^2, \qquad i \neq j$$

Together with Equation (9.1), the preceding equation shows that, for a binomial random variable X,

$$E[X^2] = np + n(n - 1)p^2$$

implying that

$$\text{Var}(X) = E[X^2] - (E[X])^2 = np + n(n - 1)p^2 - n^2p^2 = np(1 - p)$$

On the other hand, if X is hypergeometric, then, given that a white ball is chosen in trial i, each of the other $N - 1$ balls, of which $m - 1$ are white, is equally likely to be the jth ball chosen, for $j \neq i$. Consequently, for $j \neq i$,

$$P\{X_i = 1, X_j = 1\} = P\{X_i = 1\}P\{X_j = 1|X_i = 1\} = \frac{m}{N}\frac{m - 1}{N - 1}$$

Using $p_i = m/N$, we now obtain, from Equation (9.1),

$$E[X^2] = \frac{nm}{N} + n(n - 1)\frac{m}{N}\frac{m - 1}{N - 1}$$

Consequently,

$$\text{Var}(X) = \frac{nm}{N} + n(n - 1)\frac{m}{N}\frac{m - 1}{N - 1} - \left(\frac{nm}{N}\right)^2$$

which, as shown in Example 8j, can be simplified to yield

$$\text{Var}(X) = np(1 - p)\left(1 - \frac{n - 1}{N - 1}\right)$$

where $p = m/N$. ■

4.10 PROPERTIES OF THE CUMULATIVE DISTRIBUTION FUNCTION

Recall that, for the distribution function F of X, $F(b)$ denotes the probability that the random variable X takes on a value that is less than or equal to b. Following are some properties of the cumulative distribution function (c.d.f.) F:

1. F is a nondecreasing function; that is, if $a < b$, then $F(a) \leq F(b)$.
2. $\lim_{b\to\infty} F(b) = 1$.
3. $\lim_{b\to-\infty} F(b) = 0$.
4. F is right continuous. That is, for any b and any decreasing sequence $b_n, n \geq 1$, that converges to b, $\lim_{n\to\infty} F(b_n) = F(b)$.

Property 1 follows, as was noted in Section 4.1, because, for $a < b$, the event $\{X \leq a\}$ is contained in the event $\{X \leq b\}$ and so cannot have a larger probability. Properties 2, 3, and 4 all follow from the continuity property of probabilities

(Section 2.6). For instance, to prove property 2, we note that if b_n increases to ∞, then the events $\{X \leq b_n\}, n \geq 1$, are increasing events whose union is the event $\{X < \infty\}$. Hence, by the continuity property of probabilities,

$$\lim_{n\to\infty} P\{X \leq b_n\} = P\{X < \infty\} = 1$$

which proves property 2.

The proof of property 3 is similar and is left as an exercise. To prove property 4, we note that if b_n decreases to b, then $\{X \leq b_n\}, n \geq 1$, are decreasing events whose intersection is $\{X \leq b\}$. The continuity property then, yields

$$\lim_{n} P\{X \leq b_n\} = P\{X \leq b\}$$

which verifies property 4.

All probability questions about X can be answered in terms of the c.d.f., F. For example,

$$P\{a < X \leq b\} = F(b) - F(a) \quad \text{for all } a < b \tag{8.1}$$

This equation can best be seen to hold if we write the event $\{X \leq b\}$ as the union of the mutually exclusive events $\{X \leq a\}$ and $\{a < X \leq b\}$. That is,

$$\{X \leq b\} = \{X \leq a\} \cup \{a < X \leq b\}$$

so

$$P\{X \leq b\} = P\{X \leq a\} + P\{a < X \leq b\}$$

which establishes Equation (9.1).

If we want to compute the probability that X is strictly less than b, we can again apply the continuity property to obtain

$$\begin{aligned} P\{X < b\} &= P\left(\lim_{n\to\infty} \left\{X \leq b - \frac{1}{n}\right\}\right) \\ &= \lim_{n\to\infty} P\left(X \leq b - \frac{1}{n}\right) \\ &= \lim_{n\to\infty} F\left(b - \frac{1}{n}\right) \end{aligned}$$

Note that $P\{X < b\}$ does not necessarily equal $F(b)$, since $F(b)$ also includes the probability that X equals b.

EXAMPLE 10a

The distribution function of the random variable X is given by

$$F(x) = \begin{cases} 0 & x < 0 \\ \frac{x}{2} & 0 \leq x < 1 \\ \frac{2}{3} & 1 \leq x < 2 \\ \frac{11}{12} & 2 \leq x < 3 \\ 1 & 3 \leq x \end{cases}$$

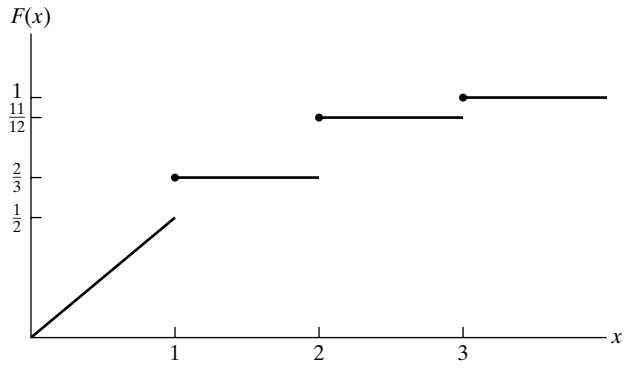


FIGURE 4.8: Graph of $F(x)$.

A graph of $F(x)$ is presented in Figure 4.8. Compute (a) $P\{X < 3\}$, (b) $P\{X = 1\}$, (c) $P\{X > \frac{1}{2}\}$, and (d) $P\{2 < X \leq 4\}$.

Solution. (a) $P\{X < 3\} = \lim_n P\left\{X \leq 3 - \frac{1}{n}\right\} = \lim_n F\left(3 - \frac{1}{n}\right) = \frac{11}{12}$

(b)

$$\begin{aligned} P\{X = 1\} &= P\{X \leq 1\} - P\{X < 1\} \\ &= F(1) - \lim_n F\left(1 - \frac{1}{n}\right) = \frac{2}{3} - \frac{1}{2} = \frac{1}{6} \end{aligned}$$

(c)

$$\begin{aligned} P\left\{X > \frac{1}{2}\right\} &= 1 - P\left\{X \leq \frac{1}{2}\right\} \\ &= 1 - F\left(\frac{1}{2}\right) = \frac{3}{4} \end{aligned}$$

(d)

$$\begin{aligned} P\{2 < X \leq 4\} &= F(4) - F(2) \\ &= \frac{1}{12} \end{aligned}$$

■

SUMMARY

A real-valued function defined on the outcome of a probability experiment is called a *random variable*.

If X is a random variable, then the function $F(x)$ defined by

$$F(x) = P\{X \leq x\}$$

is called the *distribution function* of X. All probabilities concerning X can be stated in terms of F.

A random variable whose set of possible values is either finite or countably infinite is called *discrete*. If X is a discrete random variable, then the function

$$p(x) = P\{X = x\}$$

is called the *probability mass function* of X. Also, the quantity $E[X]$ defined by

$$E[X] = \sum_{x:p(x)>0} xp(x)$$

is called the *expected value* of X. $E[X]$ is also commonly called the *mean* or the *expectation* of X.

A useful identity states that, for a function g,

$$E[g(X)] = \sum_{x:p(x)>0} g(x)p(x)$$

The *variance* of a random variable X, denoted by $\mathrm{Var}(X)$, is defined by

$$\mathrm{Var}(X) = E[(X - E[X])^2]$$

The variance, which is equal to the expected square of the difference between X and its expected value, is a measure of the spread of the possible values of X. A useful identity is

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2$$

The quantity $\sqrt{\mathrm{Var}(X)}$ is called the *standard deviation* of X.

We now note some common types of discrete random variables. The random variable X whose probability mass function is given by

$$p(i) = \binom{n}{i} p^i(1 - p)^{n-i} \quad i = 0, \ldots, n$$

is said to be a binomial random variable with parameters n and p. Such a random variable can be interpreted as being the number of successes that occur when n independent trials, each of which results in a success with probability p, are performed. Its mean and variance are given by

$$E[X] = np \quad \mathrm{Var}(X) = np(1 - p)$$

The random variable X whose probability mass function is given by

$$p(i) = \frac{e^{-\lambda}\lambda^i}{i!} \quad i \geq 0$$

is said to be a *Poisson* random variable with parameter λ. If a large number of (approximately) independent trials are performed, each having a small probability of being successful, then the number of successful trials that result will have a distribution which is approximately that of a Poisson random variable. The mean and variance of a Poisson random variable are both equal to its parameter λ. That is,

$$E[X] = \mathrm{Var}(X) = \lambda$$

The random variable X whose probability mass function is given by

$$p(i) = p(1 - p)^{i-1} \quad i = 1, 2, \ldots$$

is said to be a *geometric* random variable with parameter p. Such a random variable represents the trial number of the first success when each trial is independently a success with probability p. Its mean and variance are given by

$$E[X] = \frac{1}{p} \quad \text{Var}(X) = \frac{1 - p}{p^2}$$

The random variable X whose probability mass function is given by

$$p(i) = \binom{i-1}{r-1} p^r (1-p)^{i-r} \quad i \geq r$$

is said to be a *negative binomial* random variable with parameters r and p. Such a random variable represents the trial number of the rth success when each trial is independently a success with probability p. Its mean and variance are given by

$$E[X] = \frac{r}{p} \quad \text{Var}(X) = \frac{r(1-p)}{p^2}$$

A *hypergeometric* random variable X with parameters n, N, and m represents the number of white balls selected when n balls are randomly chosen from an urn that contains N balls of which m are white. The probability mass function of this random variable is given by

$$p(i) = \frac{\binom{m}{i}\binom{N-m}{n-i}}{\binom{N}{n}} \quad i = 0, \ldots, m$$

With $p = m/N$, its mean and variance are

$$E[X] = np \quad \text{Var}(X) = \frac{N-n}{N-1} np(1-p)$$

An important property of the expected value is that the expected value of a sum of random variables is equal to the sum of their expected values. That is,

$$E\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} E[X_i]$$

PROBLEMS

4.1. Two balls are chosen randomly from an urn containing 8 white, 4 black, and 2 orange balls. Suppose that we win \$2 for each black ball selected and we lose \$1 for each white ball selected. Let X denote our winnings. What are the possible values of X, and what are the probabilities associated with each value?

4.2. Two fair dice are rolled. Let X equal the product of the 2 dice. Compute $P\{X = i\}$ for $i = 1, \ldots, 36$.

4.3. Three dice are rolled. By assuming that each of the $6^3 = 216$ possible outcomes is equally likely, find the probabilities attached to the possible values that X can take on, where X is the sum of the 3 dice.

4.4. Five men and 5 women are ranked according to their scores on an examination. Assume that no two scores are alike and all 10! possible rankings are equally likely. Let X denote the highest ranking achieved by a woman. (For instance, $X = 1$

if the top-ranked person is female.) Find P{X = i}, $i = 1, 2, 3, \ldots, 8, 9, 10$.

4.5. Let X represent the difference between the number of heads and the number of tails obtained when a coin is tossed n times. What are the possible values of X?

4.6. In Problem 5, for $n = 3$, if the coin is assumed fair, what are the probabilities associated with the values that X can take on?

4.7. Suppose that a die is rolled twice. What are the possible values that the following random variables can take on:
- **(a)** the maximum value to appear in the two rolls;
- **(b)** the minimum value to appear in the two rolls;
- **(c)** the sum of the two rolls;
- **(d)** the value of the first roll minus the value of the second roll?

4.8. If the die in Problem 7 is assumed fair, calculate the probabilities associated with the random variables in parts (a) through (d).

4.9. Repeat Example 1b when the balls are selected with replacement.

4.10. In Example 1d, compute the conditional probability that we win i dollars, given that we win something; compute it for $i = 1, 2, 3$.

4.11. **(a)** An integer N is to be selected at random from $\{1, 2, \ldots, (10)^3\}$ in the sense that each integer has the same probability of being selected. What is the probability that N will be divisible by 3? by 5? by 7? by 15? by 105? How would your answer change if $(10)^3$ is replaced by $(10)^k$ as k became larger and larger?

(b) An important function in number theory—one whose properties can be shown to be related to what is probably the most important unsolved problem of mathematics, the Riemann hypothesis—is the Möbius function $\mu(n)$, defined for all positive integral values n as follows: Factor n into its prime factors. If there is a repeated prime factor, as in $12 = 2 \cdot 2 \cdot 3$ or $49 = 7 \cdot 7$, then $\mu(n)$ is defined to equal 0. Now let N be chosen at random from $\{1, 2, \ldots (10)^k\}$, where k is large. Determine $P\{\mu(N) = 0\}$ as $k \to \infty$.

Hint: To compute $P\{\mu(N) \neq 0\}$, use the identity

$$\prod_{i=1}^{\infty} \frac{P_i^2 - 1}{P_i^2} = \left(\frac{3}{4}\right)\left(\frac{8}{9}\right)\left(\frac{24}{25}\right)\left(\frac{48}{49}\right)\cdots = \frac{6}{\pi^2}$$

where P_i is the ith-smallest prime. (The number 1 is not a prime.)

4.12. In the game of Two-Finger Morra, 2 players show 1 or 2 fingers and simultaneously guess the number of fingers their opponent will show. If only one of the players guesses correctly, he wins an amount (in dollars) equal to the sum of the fingers shown by him and his opponent. If both players guess correctly or if neither guesses correctly, then no money is exchanged. Consider a specified player, and denote by X the amount of money he wins in a single game of Two-Finger Morra.
- **(a)** If each player acts independently of the other, and if each player makes his choice of the number of fingers he will hold up and the number he will guess that his opponent will hold up in such a way that each of the 4 possibilities is equally likely, what are the possible values of X and what are their associated probabilities?
- **(b)** Suppose that each player acts independently of the other. If each player decides to hold up the same number of fingers that he guesses his opponent will hold up, and if each player is equally likely to hold up 1 or 2 fingers, what are the possible values of X and their associated probabilities?

4.13. A salesman has scheduled two appointments to sell encyclopedias. His first appointment will lead to a sale with probability .3, and his second will lead independently to a sale with probability .6. Any sale made is equally likely to be either for the deluxe model, which costs \$1000, or the standard model, which costs \$500. Determine the probability mass function of X, the total dollar value of all sales.

4.14. Five distinct numbers are randomly distributed to players numbered 1 through 5. Whenever two players compare their numbers, the one with the higher one is declared the winner. Initially, players 1 and 2 compare their numbers; the winner then compares her number with that of player 3, and so on. Let X denote the number of times player 1 is a winner. Find $P\{X = i\}, i = 0, 1, 2, 3, 4$.

4.15. The National Basketball Association (NBA) draft lottery involves the 11 teams that had the worst won–lost records during the year. A total of 66 balls are placed in an urn. Each of these balls is inscribed with the name of a team: Eleven have the name of the team with the worst record, 10 have the name of the team with the second-worst record, 9 have the name of the team with the third-worst record, and so on (with 1 ball having the name of the team with the 11th-worst record). A ball is then chosen at random, and the team whose name is on the ball is given the first pick in the draft of players about to enter the league. Another ball is then chosen, and if it "belongs" to a team different from the one that received the first draft pick, then the team to which it belongs receives the second draft pick. (If the ball belongs

to the team receiving the first pick, then it is discarded and another one is chosen; this continues until the ball of another team is chosen.) Finally, another ball is chosen, and the team named on the ball (provided that it is different from the previous two teams) receives the third draft pick. The remaining draft picks 4 through 11 are then awarded to the 8 teams that did not "win the lottery," in inverse order of their won–lost records. For instance, if the team with the worst record did not receive any of the 3 lottery picks, then that team would receive the fourth draft pick. Let X denote the draft pick of the team with the worst record. Find the probability mass function of X.

4.16. In Problem 15, let team number 1 be the team with the worst record, let team number 2 be the team with the second-worst record, and so on. Let Y_i denote the team that gets draft pick number i. (Thus, $Y_1 = 3$ if the first ball chosen belongs to team number 3.) Find the probability mass function of (a) Y_1, (b) Y_2, and (c) Y_3.

4.17. Suppose that the distribution function of X is given by

$$F(b) = \begin{cases} 0 & b < 0 \\ \frac{b}{4} & 0 \le b < 1 \\ \frac{1}{2} + \frac{b-1}{4} & 1 \le b < 2 \\ \frac{11}{12} & 2 \le b < 3 \\ 1 & 3 \le b \end{cases}$$

(a) Find $P\{X = i\}, i = 1, 2, 3$.
(b) Find $P\{\frac{1}{2} < X < \frac{3}{2}\}$.

4.18. Four independent flips of a fair coin are made. Let X denote the number of heads obtained. Plot the probability mass function of the random variable $X - 2$.

4.19. If the distribution function of X is given by

$$F(b) = \begin{cases} 0 & b < 0 \\ \frac{1}{2} & 0 \le b < 1 \\ \frac{3}{5} & 1 \le b < 2 \\ \frac{4}{5} & 2 \le b < 3 \\ \frac{9}{10} & 3 \le b < 3.5 \\ 1 & b \ge 3.5 \end{cases}$$

calculate the probability mass function of X.

4.20. A gambling book recommends the following "winning strategy" for the game of roulette: Bet \$1 on red. If red appears (which has probability $\frac{18}{38}$), then take the \$1 profit and quit. If red does not appear and you lose this bet (which has probability $\frac{20}{38}$ of occurring), make additional \$1 bets on red on each of the next two spins of the roulette wheel and then quit. Let X denote your winnings when you quit.
(a) Find $P\{X > 0\}$.
(b) Are you convinced that the strategy is indeed a "winning" strategy? Explain your answer!
(c) Find $E[X]$.

4.21. Four buses carrying 148 students from the same school arrive at a football stadium. The buses carry, respectively, 40, 33, 25, and 50 students. One of the students is randomly selected. Let X denote the number of students that were on the bus carrying the randomly selected student. One of the 4 bus drivers is also randomly selected. Let Y denote the number of students on her bus.
(a) Which of $E[X]$ or $E[Y]$ do you think is larger? Why?
(b) Compute $E[X]$ and $E[Y]$.

4.22. Suppose that two teams play a series of games that ends when one of them has won i games. Suppose that each game played is, independently, won by team A with probability p. Find the expected number of games that are played when (a) $i = 2$ and (b) $i = 3$. Also, show in both cases that this number is maximized when $p = \frac{1}{2}$.

4.23. You have \$1000, and a certain commodity presently sells for \$2 per ounce. Suppose that after one week the commodity will sell for either \$1 or \$4 an ounce, with these two possibilities being equally likely.
(a) If your objective is to maximize the expected amount of money that you possess at the end of the week, what strategy should you employ?
(b) If your objective is to maximize the expected amount of the commodity that you possess at the end of the week, what strategy should you employ?

4.24. A and B play the following game: A writes down either number 1 or number 2, and B must guess which one. If the number that A has written down is i and B has guessed correctly, B receives i units from A. If B makes a wrong guess, B pays $\frac{3}{4}$ unit to A. If B randomizes his decision by guessing 1 with probability p and 2 with probability $1 - p$, determine his expected gain if (a) A has written down number 1 and (b) A has written down number 2.

What value of p maximizes the minimum possible value of B's expected gain, and what is this maximin value? (Note that B's expected

gain depends not only on p, but also on what A does.)

Consider now player A. Suppose that she also randomizes her decision, writing down number 1 with probability q. What is A's expected loss if (c) B chooses number 1 and (d) B chooses number 2?

What value of q minimizes A's maximum expected loss? Show that the minimum of A's maximum expected loss is equal to the maximum of B's minimum expected gain. This result, known as the minimax theorem, was first established in generality by the mathematician John von Neumann and is the fundamental result in the mathematical discipline known as the theory of games. The common value is called the value of the game to player B.

4.25. Two coins are to be flipped. The first coin will land on heads with probability .6, the second with probability .7. Assume that the results of the flips are independent, and let X equal the total number of heads that result.

(a) Find $P\{X = 1\}$.

(b) Determine $E[X]$.

4.26. One of the numbers 1 through 10 is randomly chosen. You are to try to guess the number chosen by asking questions with "yes–no" answers. Compute the expected number of questions you will need to ask in each of the following two cases:

(a) Your ith question is to be "Is it i?" $i = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10$.

(b) With each question you try to eliminate one-half of the remaining numbers, as nearly as possible.

4.27. An insurance company writes a policy to the effect that an amount of money A must be paid if some event E occurs within a year. If the company estimates that E will occur within a year with probability p, what should it charge the customer in order that its expected profit will be 10 percent of A?

4.28. A sample of 3 items is selected at random from a box containing 20 items of which 4 are defective. Find the expected number of defective items in the sample.

4.29. There are two possible causes for a breakdown of a machine. To check the first possibility would cost C_1 dollars, and, if that were the cause of the breakdown, the trouble could be repaired at a cost of R_1 dollars. Similarly, there are costs C_2 and R_2 associated with the second possibility. Let p and $1 - p$ denote, respectively, the probabilities that the breakdown is caused by the first and second possibilities. Under what conditions on $p, C_i, R_i, i = 1, 2$, should we check the first possible cause of breakdown and then the second, as opposed to reversing the checking order, so as to minimize the expected cost involved in returning the machine to working order?

Note: If the first check is negative, we must still check the other possibility.

4.30. A person tosses a fair coin until a tail appears for the first time. If the tail appears on the nth flip, the person wins 2^n dollars. Let X denote the player's winnings. Show that $E[X] = +\infty$. This problem is known as the St. Petersburg paradox.

(a) Would you be willing to pay \$1 million to play this game once?

(b) Would you be willing to pay \$1 million for each game if you could play for as long as you liked and only had to settle up when you stopped playing?

4.31. Each night different meteorologists give us the probability that it will rain the next day. To judge how well these people predict, we will score each of them as follows: If a meteorologist says that it will rain with probability p, then he or she will receive a score of

$$\begin{array}{ll} 1 - (1 - p)^2 & \text{if it does rain} \\ 1 - p^2 & \text{if it does not rain} \end{array}$$

We will then keep track of scores over a certain time span and conclude that the meteorologist with the highest average score is the best predictor of weather. Suppose now that a given meteorologist is aware of our scoring mechanism and wants to maximize his or her expected score. If this person truly believes that it will rain tomorrow with probability p^*, what value of p should he or she assert so as to maximize the expected score?

4.32. To determine whether they have a certain disease, 100 people are to have their blood tested. However, rather than testing each individual separately, it has been decided first to place the people into groups of 10. The blood samples of the 10 people in each group will be pooled and analyzed together. If the test is negative, one test will suffice for the 10 people, whereas if the test is positive, each of the 10 people will also be individually tested and, in all, 11 tests will be made on this group. Assume that the probability that a person has the disease is .1 for all people, independently of each other, and compute the expected number of tests necessary for each group. (Note that we are assuming that the pooled test will be positive if at least one person in the pool has the disease.)

4.33. A newsboy purchases papers at 10 cents and sells them at 15 cents. However, he is not allowed to return unsold papers. If his daily demand is a binomial random variable with $n = 10, p = \frac{1}{3}$, approximately how many papers should he purchase so as to maximize his expected profit?

4.34. In Example 4b, suppose that the department store incurs an additional cost of c for each unit of unmet

demand. (This type of cost is often referred to as a goodwill cost because the store loses the goodwill of those customers whose demands it cannot meet.) Compute the expected profit when the store stocks s units, and determine the value of s that maximizes the expected profit.

4.35. A box contains 5 red and 5 blue marbles. Two marbles are withdrawn randomly. If they are the same color, then you win \$1.10; if they are different colors, then you win −\$1.00. (That is, you lose \$1.00.) Calculate
(a) the expected value of the amount you win;
(b) the variance of the amount you win.

4.36. Consider Problem 22 with $i = 2$. Find the variance of the number of games played, and show that this number is maximized when $p = \frac{1}{2}$.

4.37. Find $\text{Var}(X)$ and $\text{Var}(Y)$ for X and Y as given in Problem 21.

4.38. If $E[X] = 1$ and $\text{Var}(X) = 5$, find
(a) $E[(2 + X)^2]$;
(b) $\text{Var}(4 + 3X)$.

4.39. A ball is drawn from an urn containing 3 white and 3 black balls. After the ball is drawn, it is replaced and another ball is drawn. This process goes on indefinitely. What is the probability that, of the first 4 balls drawn, exactly 2 are white?

4.40. On a multiple-choice exam with 3 possible answers for each of the 5 questions, what is the probability that a student will get 4 or more correct answers just by guessing?

4.41. A man claims to have extrasensory perception. As a test, a fair coin is flipped 10 times and the man is asked to predict the outcome in advance. He gets 7 out of 10 correct. What is the probability that he would have done at least this well if he had no ESP?

4.42. Suppose that, in flight, airplane engines will fail with probability $1 - p$, independently from engine to engine. If an airplane needs a majority of its engines operative to complete a successful flight, for what values of p is a 5-engine plane preferable to a 3-engine plane?

4.43. A communications channel transmits the digits 0 and 1. However, due to static, the digit transmitted is incorrectly received with probability .2. Suppose that we want to transmit an important message consisting of one binary digit. To reduce the chance of error, we transmit 00000 instead of 0 and 11111 instead of 1. If the receiver of the message uses "majority" decoding, what is the probability that the message will be wrong when decoded? What independence assumptions are you making?

4.44. A satellite system consists of n components and functions on any given day if at least k of the n components function on that day. On a rainy day each of the components independently functions with probability p_1, whereas on a dry day they each independently function with probability p_2. If the probability of rain tomorrow is α, what is the probability that the satellite system will function?

4.45. A student is getting ready to take an important oral examination and is concerned about the possibility of having an "on" day or an "off" day. He figures that if he has an on day, then each of his examiners will pass him, independently of each other, with probability .8, whereas if he has an off day, this probability will be reduced to .4. Suppose that the student will pass the examination if a majority of the examiners pass him. If the student feels that he is twice as likely to have an off day as he is to have an on day, should he request an examination with 3 examiners or with 5 examiners?

4.46. Suppose that it takes at least 9 votes from a 12-member jury to convict a defendant. Suppose also that the probability that a juror votes a guilty person innocent is .2, whereas the probability that the juror votes an innocent person guilty is .1. If each juror acts independently and if 65 percent of the defendants are guilty, find the probability that the jury renders a correct decision. What percentage of defendants is convicted?

4.47. In some military courts, 9 judges are appointed. However, both the prosecution and the defense attorneys are entitled to a peremptory challenge of any judge, in which case that judge is removed from the case and is not replaced. A defendant is declared guilty if the majority of judges cast votes of guilty, and he or she is declared innocent otherwise. Suppose that when the defendant is, in fact, guilty, each judge will (independently) vote guilty with probability .7, whereas when the defendant is, in fact, innocent, this probability drops to .3.
(a) What is the probability that a guilty defendant is declared guilty when there are (i) 9, (ii) 8, and (iii) 7 judges?
(b) Repeat part (a) for an innocent defendant.
(c) If the prosecution attorney does not exercise the right to a peremptory challenge of a judge, and if the defense is limited to at most two such challenges, how many challenges should the defense attorney make if he or she is 60 percent certain that the client is guilty?

4.48. It is known that diskettes produced by a certain company will be defective with probability .01, independently of each other. The company sells the diskettes in packages of size 10 and offers a money-back guarantee that at most 1 of the 10 diskettes in the package will be defective. The guarantee is that the customer can return the entire package of diskettes if he or she finds more

than one defective diskette in it. If someone buys 3 packages, what is the probability that he or she will return exactly 1 of them?

4.49. When coin 1 is flipped, it lands on heads with probability .4; when coin 2 is flipped, it lands on heads with probability .7. One of these coins is randomly chosen and flipped 10 times.
(a) What is the probability that the coin lands on heads on exactly 7 of the 10 flips?
(b) Given that the first of these ten flips lands heads, what is the conditional probability that exactly 7 of the 10 flips land on heads?

4.50. Suppose that a biased coin that lands on heads with probability p is flipped 10 times. Given that a total of 6 heads results, find the conditional probability that the first 3 outcomes are
(a) h, t, t (meaning that the first flip results in heads, the second in tails, and the third in tails);
(b) t, h, t.

4.51. The expected number of typographical errors on a page of a certain magazine is .2. What is the probability that the next page you read contains (a) 0 and (b) 2 or more typographical errors? Explain your reasoning!

4.52. The monthly worldwide average number of airplane crashes of commercial airlines is 3.5. What is the probability that there will be
(a) at least 2 such accidents in the next month;
(b) at most 1 accident in the next month?
Explain your reasoning!

4.53. Approximately 80,000 marriages took place in the state of New York last year. Estimate the probability that, for at least one of these couples,
(a) both partners were born on April 30;
(b) both partners celebrated their birthday on the same day of the year.
State your assumptions.

4.54. Suppose that the average number of cars abandoned weekly on a certain highway is 2.2. Approximate the probability that there will be
(a) no abandoned cars in the next week;
(b) at least 2 abandoned cars in the next week.

4.55. A certain typing agency employs 2 typists. The average number of errors per article is 3 when typed by the first typist and 4.2 when typed by the second. If your article is equally likely to be typed by either typist, approximate the probability that it will have no errors.

4.56. How many people are needed so that the probability that at least one of them has the same birthday as you is greater than $\frac{1}{2}$?

4.57. Suppose that the number of accidents occurring on a highway each day is a Poisson random variable with parameter $\lambda = 3$.
(a) Find the probability that 3 or more accidents occur today.
(b) Repeat part (a) under the assumption that at least 1 accident occurs today.

4.58. Compare the Poisson approximation with the correct binomial probability for the following cases:
(a) $P\{X = 2\}$ when $n = 8, p = .1$;
(b) $P\{X = 9\}$ when $n = 10, p = .95$;
(c) $P\{X = 0\}$ when $n = 10, p = .1$;
(d) $P\{X = 4\}$ when $n = 9, p = .2$.

4.59. If you buy a lottery ticket in 50 lotteries, in each of which your chance of winning a prize is $\frac{1}{100}$, what is the (approximate) probability that you will win a prize
(a) at least once?
(b) exactly once?
(c) at least twice?

4.60. The number of times that a person contracts a cold in a given year is a Poisson random variable with parameter $\lambda = 5$. Suppose that a new wonder drug (based on large quantities of vitamin C) has just been marketed that reduces the Poisson parameter to $\lambda = 3$ for 75 percent of the population. For the other 25 percent of the population, the drug has no appreciable effect on colds. If an individual tries the drug for a year and has 2 colds in that time, how likely is it that the drug is beneficial for him or her?

4.61. The probability of being dealt a full house in a hand of poker is approximately .0014. Find an approximation for the probability that, in 1000 hands of poker, you will be dealt at least 2 full houses.

4.62. Consider n independent trials, each of which results in one of the outcomes $1, \ldots, k$ with respective probabilities $p_1, \ldots, p_k$, $\sum_{i=1}^{k} p_i = 1$. Show that if all the p_i are small, then the probability that no trial outcome occurs more than once is approximately equal to $\exp(-n(n-1)\sum_i p_i^2/2)$.

4.63. People enter a gambling casino at a rate of 1 every 2 minutes.
(a) What is the probability that no one enters between 12:00 and 12:05?
(b) What is the probability that at least 4 people enter the casino during that time?

4.64. The suicide rate in a certain state is 1 suicide per 100,000 inhabitants per month.
(a) Find the probability that, in a city of 400,000 inhabitants within this state, there will be 8 or more suicides in a given month.

(b) What is the probability that there will be at least 2 months during the year that will have 8 or more suicides?

(c) Counting the present month as month number 1, what is the probability that the first month to have 8 or more suicides will be month number $i, i \geq 1$?

What assumptions are you making?

4.65. Each of 500 soldiers in an army company independently has a certain disease with probability $1/10^3$. This disease will show up in a blood test, and to facilitate matters, blood samples from all 500 soldiers are pooled and tested.

(a) What is the (approximate) probability that the blood test will be positive (that is, at least one person has the disease)?

Suppose now that the blood test yields a positive result.

(b) What is the probability, under this circumstance, that more than one person has the disease?

One of the 500 people is Jones, who knows that he has the disease.

(c) What does Jones think is the probability that more than one person has the disease?

Because the pooled test was positive, the authorities have decided to test each individual separately. The first $i - 1$ of these tests were negative, and the ith one—which was on Jones—was positive.

(d) Given the preceding, scenario, what is the probability, as a function of i, that any of the remaining people have the disease?

4.66. A total of $2n$ people, consisting of n married couples, are randomly seated (all possible orderings being equally likely) at a round table. Let C_i denote the event that the members of couple i are seated next to each other, $i = 1, \ldots, n$.

(a) Find $P(C_i)$.

(b) For $j \neq i$, find $P(C_j|C_i)$.

(c) Approximate the probability, for n large, that there are no married couples who are seated next to each other.

4.67. Repeat the preceding problem when the seating is random but subject to the constraint that the men and women alternate.

4.68. In response to an attack of 10 missiles, 500 antiballistic missiles are launched. The missile targets of the antiballistic missiles are independent, and each antiballstic missile is equally likely to go towards any of the target missiles. If each antiballistic missile independently hits its target with probability .1, use the Poisson paradigm to approximate the probability that all missiles are hit.

4.69. A fair coin is flipped 10 times. Find the probability that there is a string of 4 consecutive heads by

(a) using the formula derived in the text;

(b) using the recursive equations derived in the text.

(c) Compare your answer with that given by the Poisson approximation.

4.70. At time 0, a coin that comes up heads with probability p is flipped and falls to the ground. Suppose it lands on heads. At times chosen according to a Poisson process with rate λ, the coin is picked up and flipped. (Between these times the coin remains on the ground.) What is the probability that the coin is on its head side at time t?

Hint What would be the conditional probability if there were no additional flips by time t, and what would it be if there were additional flips by time t?

4.71. Consider a roulette wheel consisting of 38 numbers 1 through 36, 0, and double 0. If Smith always bets that the outcome will be one of the numbers 1 through 12, what is the probability that

(a) Smith will lose his first 5 bets;

(b) his first win will occur on his fourth bet?

4.72. Two athletic teams play a series of games; the first team to win 4 games is declared the overall winner. Suppose that one of the teams is stronger than the other and wins each game with probability .6, independently of the outcomes of the other games. Find the probability, for $i = 4, 5, 6, 7$, that the stronger team wins the series in exactly i games. Compare the probability that the stronger team wins with the probability that it would win a 2-out-of-3 series.

4.73. Suppose in Problem 72 that the two teams are evenly matched and each has probability $\frac{1}{2}$ of winning each game. Find the expected number of games played.

4.74. An interviewer is given a list of people she can interview. If the interviewer needs to interview 5 people, and if each person (independently) agrees to be interviewed with probability $\frac{2}{3}$, what is the probability that her list of people will enable her to obtain her necessary number of interviews if the list consists of (a) 5 people and (b) 8 people? For part (b), what is the probability that the interviewer will speak to exactly (c) 6 people and (d) 7 people on the list?

4.75. A fair coin is continually flipped until heads appears for the 10th time. Let X denote the number of tails that occur. Compute the probability mass function of X.

4.76. Solve the Banach match problem (Example 8e) when the left-hand matchbox originally contained

N_1 matches and the right-hand box contained N_2 matches.

4.77. In the Banach matchbox problem, find the probability that, at the moment when the first box is emptied (as opposed to being found empty), the other box contains exactly k matches.

4.78. An urn contains 4 white and 4 black balls. We randomly choose 4 balls. If 2 of them are white and 2 are black, we stop. If not, we replace the balls in the urn and again randomly select 4 balls. This continues until exactly 2 of the 4 chosen are white. What is the probability that we shall make exactly n selections?

4.79. Suppose that a batch of 100 items contains 6 that are defective and 94 that are not defective. If X is the number of defective items in a randomly drawn sample of 10 items from the batch, find (a) $P\{X = 0\}$ and (b) $P\{X > 2\}$.

4.80. A game popular in Nevada gambling casinos is Keno, which is played as follows: Twenty numbers are selected at random by the casino from the set of numbers 1 through 80. A player can select from 1 to 15 numbers; a win occurs if some fraction of the player's chosen subset matches any of the 20 numbers drawn by the house. The payoff is a function of the number of elements in the player's selection and the number of matches. For instance, if the player selects only 1 number, then he or she wins if this number is among the set of 20, and the payoff is \$2.2 won for every dollar bet. (As the player's probability of winning in this case is $\frac{1}{4}$, it is clear that the "fair" payoff should be \$3 won for every \$1 bet.) When the player selects 2 numbers, a payoff (of odds) of \$12 won for every \$1 bet is made when both numbers are among the 20,

(a) What would be the fair payoff in this case? Let $P_{n,k}$ denote the probability that exactly k of the n numbers chosen by the player are among the 20 selected by the house.

(b) Compute $P_{n,k}$

(c) The most typical wager at Keno consists of selecting 10 numbers. For such a bet the casino pays off as shown in the following table. Compute the expected payoff:

Keno Payoffs in 10 Number Bets	
Number of matches	Dollars won for each \$1 bet
0–4	-1
5	1
6	17
7	179
8	1,299
9	2,599
10	24,999

4.81. In Example 8i, what percentage of i defective lots does the purchaser reject? Find it for $i = 1, 4$. Given that a lot is rejected, what is the conditional probability that it contained 4 defective components?

4.82. A purchaser of transistors buys them in lots of 20. It is his policy to randomly inspect 4 components from a lot and to accept the lot only if all 4 are nondefective. If each component in a lot is, independently, defective with probability .1, what proportion of lots is rejected?

4.83. There are three highways in the county. The number of daily accidents that occur on these highways are Poisson random variables with respective parameters .3, .5, and .7. Find the expected number of accidents that will happen on any of these highways today.

4.84. Suppose that 10 balls are put into 5 boxes, with each ball independently being put in box i with probability p_i, $\sum_{i=1}^{5} p_i = 1$.

(a) Find the expected number of boxes that do not have any balls.

(b) Find the expected number of boxes that have exactly 1 ball.

4.85. There are k types of coupons. Independently of the types of previously collected coupons, each new coupon collected is of type i with probability p_i, $\sum_{i=1}^{k} p_i = 1$. If n coupons are collected, find the expected number of distinct types that appear in this set. (That is, find the expected number of types of coupons that appear at least once in the set of n coupons.)

THEORETICAL EXERCISES

4.1. There are N distinct types of coupons, and each time one is obtained it will, independently of past choices, be of type i with probability $P_i, i = 1, \ldots, N$. Let T denote the number one need select to obtain at least one of each type. Compute $P\{T = n\}$.

Hint: Use an argument similar to the one used in Example 1e.

4.2. If X has distribution function F, what is the distribution function of e^X?

4.3. If X has distribution function F, what is the distribution function of the random variable $\alpha X + \beta$, where α and β are constants, $\alpha \neq 0$?

4.4. For a nonnegative integer-valued random variable N, show that

$$E[N] = \sum_{i=1}^{\infty} P\{N \geq i\}$$

Hint: $\sum_{i=1}^{\infty} P\{N \geq i\} = \sum_{i=1}^{\infty} \sum_{k=i}^{\infty} P\{N = k\}$. Now interchange the order of summation.

4.5. For a nonnegative integer-valued random variable N, show that

$$\sum_{i=0}^{\infty} iP\{N > i\} = \frac{1}{2}(E[N^2] - E[N])$$

Hint: $\sum_{i=0}^{\infty} iP\{N > i\} = \sum_{i=0}^{\infty} i \sum_{k=i+1}^{\infty} P\{N = k\}$. Now interchange the order of summation.

4.6. Let X be such that

$$P\{X = 1\} = p = 1 - P\{X = -1\}$$

Find $c \neq 1$ such that $E[c^X] = 1$.

4.7. Let X be a random variable having expected value μ and variance σ^2. Find the expected value and variance of

$$Y = \frac{X - \mu}{\sigma}$$

4.8. Find $\text{Var}(X)$ if

$$P(X = a) = p = 1 - P(X = b)$$

4.9. Show how the derivation of the binomial probabilities

$$P\{X = i\} = \binom{n}{i} p^i (1 - p)^{n-i}, \quad i = 0, \ldots, n$$

leads to a proof of the binomial theorem

$$(x + y)^n = \sum_{i=0}^{n} \binom{n}{i} x^i y^{n-i}$$

when x and y are nonnegative.
Hint: Let $p = \frac{x}{x+y}$.

4.10. Let X be a binomial random variable with parameters n and p. Show that

$$E\left[\frac{1}{X + 1}\right] = \frac{1 - (1 - p)^{n+1}}{(n + 1)p}$$

4.11. Consider n independent sequential trials, each of which is successful with probability p. If there is a total of k successes, show that each of the $n!/[k!(n - k)!]$ possible arrangements of the k successes and $n - k$ failures is equally likely.

4.12. There are n components lined up in a linear arrangement. Suppose that each component independently functions with probability p. What is the probability that no 2 neighboring components are both nonfunctional?
Hint: Condition on the number of defective components and use the results of Example 4c of Chapter 1.

4.13. Let X be a binomial random variable with parameters (n, p). What value of p maximizes $P\{X = k\}, k = 0, 1, \ldots, n$? This is an example of a statistical method used to estimate p when a binomial (n, p) random variable is observed to equal k. If we assume that n is known, then we estimate p by choosing that value of p which maximizes $P\{X = k\}$. This is known as the *method of maximum likelihood estimation*.

4.14. A family has n children with probability $\alpha p^n, n \geq 1$, where $\alpha \leq (1 - p)/p$.
(a) What proportion of families has no children?
(b) If each child is equally likely to be a boy or a girl (independently of each other), what proportion of families consists of k boys (and any number of girls)?

4.15. Suppose that n independent tosses of a coin having probability p of coming up heads are made. Show that the probability that an even number of heads results is $\frac{1}{2}[1 + (q - p)^n]$, where $q = 1 - p$. Do this by proving and then utilizing the identity

$$\sum_{i=0}^{[n/2]} \binom{n}{2i} p^{2i} q^{n-2i} = \frac{1}{2}\left[(p + q)^n + (q - p)^n\right]$$

where $[n/2]$ is the largest integer less than or equal to $n/2$. Compare this exercise with Theoretical Exercise 3.5 of Chapter 3.

4.16. Let X be a Poisson random variable with parameter λ. Show that $P\{X = i\}$ increases monotonically and then decreases monotonically as i increases, reaching its maximum when i is the largest integer not exceeding λ.
Hint: Consider $P\{X = i\}/P\{X = i - 1\}$.

4.17. Let X be a Poisson random variable with parameter λ.
(a) Show that

$$P\{X \text{ is even}\} = \frac{1}{2}\left[1 + e^{-2\lambda}\right]$$

by using the result of Theoretical Exercise 15 and the relationship between Poisson and binomial random variables.

(b) Verify the formula in part (a) directly by making use of the expansion of $e^{-\lambda} + e^{\lambda}$.

4.18. Let X be a Poisson random variable with parameter λ. What value of λ maximizes $P\{X = k\}, k \geq 0$?

4.19. Show that X is a Poisson random variable with parameter λ, then

$$E[X^n] = \lambda E[(X + 1)^{n-1}]$$

Now use this result to compute $E[X^3]$.

4.20. Consider n coins, each of which independently comes up heads with probability p. Suppose that n is large and p is small, and let $\lambda = np$. Suppose that all n coins are tossed; if at least one comes up heads, the experiment ends; if not, we again toss all n coins, and so on. That is, we stop the first time that at least one of the n coins come up heads. Let X denote the total number of heads that appear. Which of the following reasonings concerned with approximating $P\{X = 1\}$ is correct (in all cases, Y is a Poisson random variable with parameter λ)?

(a) Because the total number of heads that occur when all n coins are rolled is approximately a Poisson random variable with parameter λ,

$$P\{X = 1\} \approx P\{Y = 1\} = \lambda e^{-\lambda}$$

(b) Because the total number of heads that occur when all n coins are rolled is approximately a Poisson random variable with parameter λ, and because we stop only when this number is positive,

$$P\{X = 1\} \approx P\{Y = 1 | Y > 0\} = \frac{\lambda e^{-\lambda}}{1 - e^{-\lambda}}$$

(c) Because at least one coin comes up heads, X will equal 1 if none of the other $n - 1$ coins come up heads. Because the number of heads resulting from these $n - 1$ coins is approximately Poisson with mean $(n - 1)p \approx \lambda$,

$$P\{X = 1\} \approx P\{Y = 0\} = e^{-\lambda}$$

4.21. From a set of n randomly chosen people, let E_{ij} denote the event that persons i and j have the same birthday. Assume that each person is equally likely to have any of the 365 days of the year as his or her birthday. Find

(a) $P(E_{3,4}|E_{1,2})$;

(b) $P(E_{1,3}|E_{1,2})$;

(c) $P(E_{2,3}|E_{1,2} \cap E_{1,3})$.

What can you conclude from your answers to parts (a)–(c) about the independence of the $\binom{n}{2}$ events E_{ij}?

4.22. An urn contains $2n$ balls, of which 2 are numbered 1, 2 are numbered $2, \ldots,$ and 2 are numbered n. Balls are successively withdrawn 2 at a time without replacement. Let T denote the first selection in which the balls withdrawn have the same number (and let it equal infinity if none of the pairs withdrawn has the same number). We want to show that, for $0 < \alpha < 1$,

$$\lim_n P\{T > \alpha n\} = e^{-\alpha/2}$$

To verify the preceding formula, let M_k denote the number of pairs withdrawn in the first k selections, $k = 1, \ldots, n$.

(a) Argue that when n is large, M_k can be regarded as the number of successes in k (approximately) independent trials.

(b) Approximate $P\{M_k = 0\}$ when n is large.

(c) Write the event $\{T > \alpha n\}$ in terms of the value of one of the variables M_k.

(d) Verify the limiting probability given for $P\{T > \alpha n\}$.

4.23. Consider a random collection of n individuals. In approximating the probability that no 3 of these individuals share the same birthday, a better Poisson approximation than that obtained in the text (at least for values of n between 80 and 90) is obtained by letting E_i be the event that there are at least 3 birthdays on day i, $i = 1, \ldots, 365$.

(a) Find $P(E_i)$.

(b) Give an approximation for the probability that no 3 individuals share the same birthday.

(c) Evaluate the preceding when $n = 88$ (which can be shown to be the smallest value of n for which the probability exceeds .5).

4.24. Here is another way to obtain a set of recursive equations for determining P_n, the probability that there is a string of k consecutive heads in a sequence of n flips of a fair coin that comes up heads with probability p:

(a) Argue that, for $k < n$, there will be a string of k consecutive heads if either

1. there is a string of k consecutive heads within the first $n - 1$ flips, or
2. there is no string of k consecutive heads within the first $n - k - 1$ flips, flip $n - k$ is a tail, and flips $n - k + 1, \ldots, n$ are all heads.

(b) Using the preceding, relate P_n to P_{n-1}. Starting with $P_k = p^k$, the recursion can be used to obtain P_{k+1}, then P_{k+1}, and so on, up to P_n.

4.25. Suppose that the number of events that occur in a specified time is a Poisson random variable with parameter λ. If each event is counted with probability p, independently of every other event, show that the number of events that are counted is a Poisson random variable with parameter λp. Also, give an intuitive argument as to why this should be so. As an application of the preceding result, suppose that the number of distinct uranium deposits in a given area is a Poisson random variable with parameter $\lambda = 10$. If, in a fixed period of time, each deposit is discovered independently with probability $\frac{1}{50}$, find the probability that (a) exactly 1, (b) at least 1, and (c) at most 1 deposit is discovered during that time.

4.26. Prove

$$\sum_{i=0}^{n} e^{-\lambda}\frac{\lambda^i}{i!} = \frac{1}{n!}\int_{\lambda}^{\infty} e^{-x}x^n dx$$

Hint: Use integration by parts.

4.27. If X is a geometric random variable, show analytically that

$$P\{X = n + k | X > n\} = P\{X = k\}$$

Using the interpretation of a geometric random variable, give a verbal argument as to why the preceding equation is true.

4.28. Let X be a negative binomial random variable with parameters r and p, and let Y be a binomial random variable with parameters n and p. Show that

$$P\{X > n\} = P\{Y < r\}$$

Hint: Either one could attempt an analytical proof of the preceding equation, which is equivalent to proving the identity

$$\sum_{i=n+1}^{\infty}\binom{i-1}{r-1}p^r(1-p)^{i-r} = \sum_{i=0}^{r-1}\binom{n}{i} \times p^i(1-p)^{n-i}$$

or one could attempt a proof that uses the probabilistic interpretation of these random variables. That is, in the latter case, start by considering a sequence of independent trials having a common probability p of success. Then try to express the events $\{X > n\}$ and $\{Y < r\}$ in terms of the outcomes of this sequence.

4.29. For a hypergeometric random variable, determine

$$P\{X = k + 1\}/P\{X = k\}$$

4.30. Balls numbered 1 through N are in an urn. Suppose that $n, n \leq N$, of them are randomly selected without replacement. Let Y denote the largest number selected.

(a) Find the probability mass function of Y.

(b) Derive an expression for $E[Y]$ and then use Fermat's combinatorial identity (see Theoretical Exercise 11 of Chapter 1) to simplify the expression.

4.31. A jar contains $m + n$ chips, numbered $1, 2, \ldots, n + m$. A set of size n is drawn. If we let X denote the number of chips drawn having numbers that exceed each of the numbers of those remaining, compute the probability mass function of X.

4.32. A jar contains n chips. Suppose that a boy successively draws a chip from the jar, each time replacing the one drawn before drawing another. The process continues until the boy draws a chip that he has previously drawn. Let X denote the number of draws, and compute its probability mass function.

4.33. Show that Equation (8.6) follows from Equation (8.5).

4.34. From a set of n elements, a nonempty subset is chosen at random in the sense that all of the nonempty subsets are equally likely to be selected. Let X denote the number of elements in the chosen subset. Using the identities given in Theoretical Exercise 12 of Chapter 1, show that

$$E[X] = \frac{n}{2 - \left(\frac{1}{2}\right)^{n-1}}$$

$$\mathrm{Var}(X) = \frac{n \cdot 2^{2n-2} - n(n+1)2^{n-2}}{(2^n - 1)^2}$$

Show also that, for n large,

$$\mathrm{Var}(X) \backsim \frac{n}{4}$$

in the sense that the ratio $\mathrm{Var}(X)$ to $n/4$ approaches 1 as n approaches ∞. Compare this formula with the limiting form of $\mathrm{Var}(Y)$ when $P\{Y = i\} = 1/n, i = 1, \ldots, n$.

4.35. An urn initially contains one red and one blue ball. At each stage, a ball is randomly chosen and then replaced along with another of the same color. Let X denote the selection number of the first chosen ball that is blue. For instance, if the first selection is red and the second blue, then X is equal to 2.

(a) Find $P\{X > i\}, i \geq 1$.

(b) Show that, with probability 1, a blue ball is eventually chosen. (That is, show that $P\{X < \infty\} = 1$.)

(c) Find $E[X]$.

4.36. Suppose the possible values of X are $\{x_i\}$, the possible values of Y are $\{y_j\}$, and the possible values of $X + Y$ are $\{z_k\}$. Let A_k denote the set of all pairs of indices (i,j) such that $x_i + y_j = z_k$; that is, $A_k = \{(i,j) : x_i + y_j = z_k\}$.

(a) Argue that

$$P\{X + Y = z_k\} = \sum_{(i,j)\in A_k} P\{X = x_i, Y = y_j\}$$

(b) Show that

$$E[X + Y] = \sum_k \sum_{(i,j)\in A_k} (x_i + y_j)P\{X = x_i, Y = y_j\}$$

(c) Using the formula from part (b), argue that

$$E[X + Y] = \sum_i \sum_j (x_i + y_j)P\{X = x_i, Y = y_j\}$$

(d) Show that

$$P(X = x_i) = \sum_j P(X = x_i, Y = y_j),$$
$$P(Y = y_j) = \sum_i P\{X = x_i, Y = y_j\}$$

(e) Prove that

$$E[X + Y] = E[X] + E[Y]$$

SELF-TEST PROBLEMS AND EXERCISES

4.1. Suppose that the random variable X is equal to the number of hits obtained by a certain baseball player in his next 3 at bats. If $P\{X = 1\} = .3, P\{X = 2\} = .2$, and $P\{X = 0\} = 3P\{X = 3\}$, find $E[X]$.

4.2. Suppose that X takes on one of the values 0, 1, and 2. If for some constant $c, P\{X = i\} = cP\{X = i - 1\}, i = 1, 2$, find $E[X]$.

4.3. A coin that, when flipped, comes up heads with probability p is flipped until either heads or tails has occurred twice. Find the expected number of flips.

4.4. A certain community is composed of m families, n_i of which have i children, $\sum_{i=1}^{r} n_i = m$. If one of the families is randomly chosen, let X denote the number of children in that family. If one of the $\sum_{i=1}^{r} in_i$ children is randomly chosen, let Y denote the total number of children in the family of that child. Show that $E[Y] \geq E[X]$.

4.5. Suppose that $P\{X = 0\} = 1 - P\{X = 1\}$. If $E[X] = 3\text{Var}(X)$, find $P\{X = 0\}$.

4.6. There are 2 coins in a bin. When one of them is flipped, it lands on heads with probability .6, and when the other is flipped, it lands on heads with probability .3. One of these coins is to be randomly chosen and then flipped. Without knowing which coin is chosen, you can bet any amount up to 10 dollars, and you then either win that amount if the coin comes up heads or lose it if it comes up tails. Suppose, however, that an insider is willing to sell you, for an amount C, the information as to which coin was selected. What is your expected payoff if you buy this information? Note that if you buy it and then bet x, you will end up either winning $x - C$ or $-x - C$ (that is, losing $x + C$ in the latter case). Also, for what values of C does it pay to purchase the information?

4.7. A philanthropist writes a positive number x on a piece of red paper, shows the paper to an impartial observer, and then turns it face down on the table. The observer then flips a fair coin. If it shows heads, she writes the value $2x$ and, if tails, the value $x/2$, on a piece of blue paper, which she then turns face down on the table. Without knowing either the value x or the result of the coin flip, you have the option of turning over either the red or the blue piece of paper. After doing so and observing the number written on that paper, you may elect to receive as a reward either that amount or the (unknown) amount written on the other piece of paper. For instance, if you elect to turn over the blue paper and observe the value 100, then you can elect either to accept 100 as your reward or to take the amount (either 200 or 50) on the red paper. Suppose that you would like your expected reward to be large.

(a) Argue that there is no reason to turn over the red paper first, because if you do so, then no matter what value you observe, it is always better to switch to the blue paper.

(b) Let y be a fixed nonnegative value, and consider the following strategy: Turn over the blue paper, and if its value is at least y, then accept that amount. If it is less than y, then switch to the red paper. Let $R_y(x)$ denote the reward obtained if the philanthropist writes the amount x and you employ this strategy. Find $E[R_y(x)]$. Note that $E[R_0(x)]$ is the

expected reward if the philanthropist writes the amount x when you employ the strategy of always choosing the blue paper.

4.8. Let $B(n, p)$ represent a binomial random variable with parameters n and p. Argue that

$$P\{B(n,p) \le i\} = 1 - P\{B(n,1-p) \le n - i - 1\}$$

Hint: The number of successes less than or equal to i is equivalent to what statement about the number of failures?

4.9. If X is a binomial random variable with expected value 6 and variance 2.4, find $P\{X = 5\}$.

4.10. An urn contains n balls numbered 1 through n. If you withdraw m balls randomly in sequence, each time replacing the ball selected previously, find $P\{X = k\}, k = 1, \ldots, m$, where X is the maximum of the m chosen numbers.
Hint: First find $P\{X \le k\}$.

4.11. Teams A and B play a series of games, with the first team to win 3 games being declared the winner of the series. Suppose that team A independently wins each game with probability p. Find the conditional probability that team A wins
(a) the series given that it wins the first game;
(b) the first game given that it wins the series.

4.12. A local soccer team has 5 more games left to play. If it wins its game this weekend, then it will play its final 4 games in the upper bracket of its league, and if it loses, then it will play its final games in the lower bracket. If it plays in the upper bracket, then it will independently win each of its games in this bracket with probability .4, and if it plays in the lower bracket, then it will independently win each of its games with probability .7. If the probability that the team wins its game this weekend is .5, what is the probability that it wins at least 3 of its final 4 games?

4.13. Each of the members of a 7-judge panel independently makes a correct decision with probability .7. If the panel's decision is made by majority rule, what is the probability that the panel makes the correct decision? Given that 4 of the judges agreed, what is the probability that the panel made the correct decision?

4.14. On average, 5.2 hurricanes hit a certain region in a year. What is the probability that there will be 3 or fewer hurricanes hitting this year?

4.15. The number of eggs laid on a tree leaf by an insect of a certain type is a Poisson random variable with parameter λ. However, such a random variable can be observed only if it is positive, since if it is 0 then we cannot know that such an insect was on the leaf. If we let Y denote the observed number of eggs, then

$$P\{Y = i\} = P\{X = i | X > 0\}$$

where X is Poisson with parameter λ. Find $E[Y]$.

4.16. Each of n boys and n girls, independently and randomly, chooses a member of the other sex. If a boy and girl choose each other, they become a couple. Number the girls, and let G_i be the event that girl number i is part of a couple. Let $P_0 = 1 - P(\cup_{i=1}^{n} G_i)$ be the probability that no couples are formed.
(a) What is $P(G_i)$?
(b) What is $P(G_i|G_j)$?
(c) When n is large, approximate P_0.
(d) When n is large, approximate P_k, the probability that exactly k couples are formed.
(e) Use the inclusion–exclusion identity to evaluate P_0.

4.17. A total of $2n$ people, consisting of n married couples, are randomly divided into n pairs. Arbitrarily number the women, and let W_i denote the event that woman i is paired with her husband.
(a) Find $P(W_i)$.
(b) For $i \ne j$, find $P(W_i|W_j)$.
(c) When n is large, approximate the probability that no wife is paired with her husband.
(d) If each pairing must consist of a man and a woman, what does the problem reduce to?

4.18. A casino patron will continue to make \$5 bets on red in roulette until she has won 4 of these bets.
(a) What is the probability that she places a total of 9 bets?
(b) What is her expected winnings when she stops?
Remark: On each bet, she will either win \$5 with probability $\frac{18}{38}$ or lose \$5 with probability $\frac{20}{38}$.

4.19. When three friends go for coffee, they decide who will pay the check by each flipping a coin and then letting the "odd person" pay. If all three flips produce the same result (so that there is no odd person), then they make a second round of flips, and they continue to do so until there is an odd person. What is the probability that
(a) exactly 3 rounds of flips are made?
(b) more than 4 rounds are needed?

4.20. Show that if X is a geometric random variable with parameter p, then

$$E[1/X] = \frac{-p\log(p)}{1-p}$$

Hint: You will need to evaluate an expression of the form $\sum_{i=1}^{\infty} a^i/i$. To do so, write $a^i/i = \int_0^a x^{i-1}dx$, and then interchange the sum and the integral.

4.21. Suppose that

$$P\{X = a\} = p, \quad P\{X = b\} = 1 - p$$

(a) Show that $\frac{X-b}{a-b}$ is a Bernoulli random variable.
(b) Find Var(X).

4.22. Each game you play is a win with probability p. You plan to play 5 games, but if you win the fifth game, then you will keep on playing until you lose.
(a) Find the expected number of games that you play.
(b) Find the expected number of games that you lose.

4.23. Balls are randomly withdrawn, one at a time without replacement, from an urn that initially has N white and M black balls. Find the probability that n white balls are drawn before m black balls, $n \leq N, m \leq M$.

4.24. Ten balls are to be distributed among 5 urns, with each ball going into urn i with probability p_i, $\sum_{i=1}^{5} p_i = 1$. Let X_i denote the number of balls that go into urn i. Assume that events corresponding to the locations of different balls are independent.
(a) What type of random variable is X_i? Be as specific as possible.
(b) For $i \neq j$, what type of random variable is $X_i + X_j$?
(c) Find $P\{X_1 + X_2 + X_3 = 7\}$.

4.25. For the match problem (Example 5m in Chapter 2), find
(a) the expected number of matches.
(b) the variance of the number of matches.

4.26. Let α be the probability that a geometric random variable X with parameter p is an even number.
(a) Find α by using the identity $\alpha = \sum_{i=1}^{\infty} \mathrm{P\{X = 2i\}}$.
(b) Find α by conditioning on whether $X = 1$ or $X > 1$.